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# Emission limits and emission allocation schemes in intermodal freight transportation

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ABSTRACT

The problem addressed in this paper seeks for an optimal routing of freight orders through an intermodal transportation network. We consider the case of environmentally aware customers that request to ship orders with no more than a specified amount of greenhouse gases, which establishes so-called emission limits. In order to ensure that a routing plan complies with each order's emission limit, it is necessary to estimate emissions caused by the used transport services and to allocate these emissions to the orders. We model this problem under cost-, emission- and service-objectives and apply it to an intermodal rail/road network in Europe.

## 1. Introduction

Within the last years, an increasing environmental awareness in different areas of modern societies can be observed. One form of this development are eco-labels that indicate the energy consumption or the carbon footprint of products or services (Thøgersen and Nielsen, 2016). It is not surprising that such overall environmental cautiousness is carried over to the decision-making processes in transportation and logistics companies (e.g., DHL, 2019; UPS, 2019).

In this study, we model a service selection problem (SSP) where a set of orders, which is placed by customers, is to be routed through an intermodal network of road and rail transportation services. Thereby, we consider the case of environmentally aware customers that request to ship orders with no more than a specified amount of greenhouse gases, so called *emission limits*. The planning task is (i) to come up with a routing plan, (ii) to estimate the corresponding emissions, and (iii) to implement emission allocation schemes such that the solution complies with each order's emission limit. The routing plan indicates the services used as well as the transshipment processes required between the services. Overall, the problem addressed in this paper covers two types of decisions. Firstly, the decision about the routing and transshipment of the orders and, secondly, the decision about each service's emission allocation scheme. Since the orders are typically placed by customers and the services are typically operated by one or several companies, these decisions might be made by different actors. However, there are also settings where one decision maker is in charge of making both decisions. For example, a freight agent could act between the customers and the service operators, selecting both the services to be used and the emission allocation schemes that fit to the customers' environmental preferences. The freight agent then prescribes the required schemes for the service operators (e.g., if the agent possesses the market power) or uses this information for negotiations with the service operators. For this reason, we consider a model where both decisions are made holistically by one decision maker. Throughout this study, we consider an *egalitarian allocation* scheme and a *payload-based allocation* scheme as alternatives for emission allocation. With these schemes, we allocate emissions from a transportation service, such as a truck or a train, to orders that use this service. In particular, the egalitarian scheme allocates the same amount of emissions to each

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order of a service and the payload-based scheme allocates emissions to orders based on the orders' contribution to a service's overall payload. For reasons of comparison, we also present results for the case where all services consistently use the egalitarian allocation scheme and for the case where all services consistently use the payload-based allocation scheme. Note that by letting the model select the services' allocation schemes, an order's route might involve services that use different schemes. This helps in respecting an order's emission limit. For example, if an order exceeds its emission limit in a routing-solution under a consistent application of a single allocation scheme, then, by changing the allocation scheme on some of the used services, the emissions assigned to this particular order could fall below the emission limit. The freedom to apply different allocation schemes on different services is in accordance with the European norm on the calculation and declaration of emissions of transport services (DIN EN 16258, 2012). Overall, the **S**ervice **S**election **P**roblem in our study considers **T**ransshipment processes, **A**llocation schemes per service and **L**imits on the emissions per order in an **E**mission-oriented setting. We refer to this problem as **SSP-TALE** and formulate it for cost-, emission- and service-objectives.

With this study, we are, to the best of our knowledge, the first to present the problem of routing orders with order-specific emission limits through an intermodal network of transportation services where the emission allocation schemes are selected for each service. We show how to model and solve the corresponding problem. We analyze the impact of orders' emission limits, which serve as an indicator for customers' environmental cautiousness, on the routing of individual orders as well as their impact on the overall network costs and emissions. In addition, we show how the choice of emission allocation schemes can change the routing plan and, thus, the optimal solution under each of the considered objectives. Finally, we also present insights on the settings in which the model is likely to select either the egalitarian or the payload-based emission allocation scheme for the services in the considered network.

The rest of this paper is organized as follows. Section 2 presents an overview of relevant literature. Section 3 describes the SSP-TALE and its mathematical formulation. Section 4 presents extensive experimental results that are based on real-world data. In particular, we consider the routing of orders between Portugal and the Netherlands and between Italy and Denmark in the European intermodal rail/road network. Section 5 concludes this paper.

## 2. Literature review

In this section, we provide an overview of studies on environmentally oriented routing in transportation networks. We mainly focus on studies that address intermodal freight networks consisting of road and rail services. We also include some relevant unimodal studies that address the allocation of emissions to orders. Table 1 shows an overview of all considered studies, together with the central problem features that are relevant in relation to our study. A more general overview of studies concerning multimodal freight transportation is provided by SteadieSeifi et al. (2014).

Bauer et al. (2010) extend a service network design problem by considering a minimization of total emissions emitted by the services in the network as well as the more traditional objective of minimizing total transportation time of orders. The authors perform a computational study for a rail/road network that covers Austria, the Czech Republic and Poland and compare solutions for both objectives. In contrast to the time minimizing solution, the emission minimizing solution results in fewer used services and higher waiting times for transshipping orders between services. Rudi et al. (2016) formulate a capacitated multi-commodity network

**Table 1**
Overview of selected relevant literature.

| | | | objective | | | emissions | | | emission limits | |
|---|---|---|---|---|---|---|---|---|---|---|
| reference | multimodal | unimodal | costs | emissions | service level/ time | transportation | transshipment | allocation selection[a] | network wide | per order |
| Bauer et al. (2010) | √ | – | – | √ | √ | √ | – | – | – | – |
| Rudi et al. (2016) | √ | – | √ | √ | √ | √ | √ | – | – | – |
| Sun and Lang (2015) | √ | – | √ | √ | – | √ | – | – | – | – |
| Qu et al. (2016) | √ | – | √ | √ | – | √ | – | – | – | – |
| Baykasoğlu and Subulan (2016) | √ | – | √ | √ | √ | √ | – | – | – | – |
| Demir et al. (2016) | √ | – | √ | √ | √ | √ | √ | – | – | – |
| Sharma and Mathew (2011) | √ | – | – | √ | √ | √ | – | – | – | – |
| Heinold and Meisel (2018) | √ | – | – | √ | – | √ | √ | – | – | – |
| Heinold and Meisel (2019) | √ | – | – | √ | √ | √ | √ | – | – | – |
| Lam and Gu (2016) | √ | – | √ | – | √ | √ | – | – | √ | – |
| Kirschstein and Bierwirth (2018) | – | √ | – | √ | – | √ | – | √ | – | – |
| Kellner and Schneiderbauer (2019) | – | √ | – | √ | – | √ | – | √ | – | – |
| this study | √ | – | √ | √ | √ | √ | √ | √ | (√) | √ |

[a] studies that use multiple schemes for allocating emissions to orders.

flow model with the objectives to minimize costs, transit times and emissions of the transport and transshipment processes. The model considers in-transit inventory and finds an optimal solution for the routing and transshipment of products. Sun and Lang (2015) consider the problem of routing commodities through a rail/road transportation network. The authors consider schedule-based rail services and time-flexible road services and apply their model to a large-scale problem of bringing containers from inland cities to seaports in China. The objective is to minimize total costs for transportation, transshipment, inventory, pick-up and delivery as well as $CO_2$ emissions from transportation. Qu et al. (2016) present a similar problem and consider services' fixed and variable costs, transshipment costs as well as emissions. Here, the authors also consider transshipment costs that occur if an order has to change between one mode of transportation to another (e.g., from road to rail). Baykasoğlu and Subulan (2016) consider a multi-modal transportation problem with the objectives to minimize transport costs, transit time or transport-related emissions. The authors consider a multi-period scenario and test their model on data from an international logistics company in Turkey. Demir et al. (2016) describe a green intermodal service network design problem with travel time uncertainty. The objective of the model is to minimize a weighted cost function that considers direct transportation, transit time, and emissions. Thereby, the authors consider emissions from transportation and transshipment processes. A sample average approximation algorithm is used to find robust solutions under uncertain travel times. Sharma and Mathew (2011) consider as objectives the minimization of transit time and transport emissions in a multi-objective network design problem with various road modes. The authors formulate the problem as a bi-level model with an upper level decision on the road network design and a lower level decision on the routing in this network. Heinold and Meisel (2018) provide a comprehensive overview of emission rates for European rail/road intermodal and road-only transportation. For this, the authors conduct a systematic simulation study for the rail/road Trans-European Transport Network where they consider emissions from transportation and transshipment processes. In a follow-up study, Heinold and Meisel (2019) propose a model to route an individual order through an intermodal transportation network with the objective to minimize total emissions or transit time. The authors measure emissions and transit time from transportation and transshipment processes under both objectives and present comprehensive results for a large number of artificial orders that are shipped across Europe. Lam and Gu (2016) present a model to find an optimal routing for shipping containers in a port hinterland by using trucks, trains, and barges. A bi-objective optimization approach is used to demonstrate the trade-offs between costs and transit time. The model considers the environmental impact of a solution by restricting the total amount of emitted greenhouse gases through a hard constraint. The authors argue that such limit could be imposed by governmental or regulatory authorities. Chen et al. (2014) consider emission reduction targets in an intermodal waterway-road network. The authors provide a model that determines a routing for the orders under the objective of minimizing state subsidies. Thereby, restrictions enforce certain carbon emission reduction targets, such as reducing network emissions by at least 15%. Similarly, Chen and Wang (2016) analyze the selection of transport modes under different emission reduction policies. In particular, the authors consider the cap policy and the cap-and-trade policy that both set an overall limit for the network emissions. Anyhow, the latter two studies are only little related to our research as they consider the problem from the point of view of transport policy makers whereas our paper considers the problem from the point of view of a logistics service provider.

The previously mentioned studies apply to networks with multiple modes of transportation. Thereby, total emissions are either considered as an objective to minimize or as a limit on the overall network emissions. Studies that incorporate the selection of emission allocation schemes into routing decisions so far consider only one transportation mode. For example, Kirschstein and Bierwirth (2018) propose a road-only selective traveling salesman problem with emission allocation schemes. The problem is to find a vehicle route that minimizes the amount of greenhouse gases that is allocated to one specific order. For this purpose, the model also decides on the emission allocation scheme to use. Here, either a distance based allocation or an egalitarian allocation can be selected to distribute the total route-emissions among the orders. Kellner and Schneiderbauer (2019) further analyze the effect of allocation schemes in road transportation. The authors consider three problem types: a vehicle routing problem (VRP), a road-only network flow model (NFM), and a mixed VRP-NFM model. The authors apply various allocation schemes, including schemes that are based on principles of game theory, to each considered problem type. The authors show by experiment that emission allocation schemes that are based on the orders' individual travel distances lead to emission allocations that are similar to an allocation based on Shapley values, which serves as a benchmark allocation method. Note that Kirschstein and Bierwirth (2018) and Kellner and Schneiderbauer (2019) both analyze schemes that allocate total network emissions to individual orders. In contrast, we consider this in more detail for a situation where emission allocation schemes are selected for each service, rather than applying one scheme to the overall network. With this, it is possible to use different emission allocation schemes for the services within the considered transportation network. Accordingly, we do not consider allocation schemes that use the Shapley value or the travel distance of the orders, as these imply using the same allocation scheme for all services. Instead, we concentrate on the service-specific egalitarian scheme and payload-based scheme.

In contrast to the existing literature, the problem addressed in this paper is to route orders with *individual emission limits* through an intermodal network where we consider emissions from transportation and transshipment processes and also *select allocation schemes* to allocate these emissions to orders. The emission limits reflect individual customer preferences on the environmental impact of their shipment. As shown above, hard emission limits on a network level have been considered in some studies but no study considered emission limits per order so far. We fill this gap by presenting a model that considers emission limits per order, which, through the sum over all orders' emission limits, also imposes an overall network limit. We implement such limits as soft constraints and use the number of orders that can be routed within their emission limit as a service-oriented performance measure. We do not consider time as an additional service-oriented measure although some of the above studies include this in their analysis. This is because the focus of our study is on describing interdependencies of orders' emission limits, orders' routings, and services' emission allocation schemes. We refer to Heinold and Meisel (2019) for a study that compares the emission-oriented and the time-oriented routing of orders in an intermodal network. Furthermore, although we formulate the model as an intermodal transportation problem, it can also be applied to settings with only one mode of transportation, simply by restricting the set of considered services. In our

**Table 2**
Sets, parameters and decision variables in the SSP-TALE.

| name | description |
|---|---|
| nodes | |
| $\mathcal{N}$ | set of nodes ($\mathcal{N} = \bigcup_{o\in O}\{a_o, b_o\} \cup \bigcup_{s\in\mathcal{S}}\{a_s, b_s\}$) |
| $r_i^{tship}$ | costs in \$/TEU caused by transshipping one TEU at node $i \in \mathcal{N}$ |
| $e_i^{tship}$ | emissions in $kgCO_2e$/TEU caused by transshipping one TEU at node $i \in \mathcal{N}$ |
| orders | |
| $O$ | set of orders |
| $q_o$ | payload of order $o \in O$ (in TEU) |
| $a_o$, $b_o$ | origin/destination location (node) of order $o \in O$, $a_o, b_o \in \mathcal{N}$ |
| $\bar{e}_o$ | emission limit of order $o \in O$ (in $kgCO_2e$) |
| services | |
| $\mathcal{S}$ | set of services |
| $a_s$, $b_s$ | origin/destination location (node) of service $s \in \mathcal{S}$, $a_s, b_s \in \mathcal{N}$ |
| $suc_s$ | succeeding service of service $s \in \mathcal{S}$, $suc_s \in \mathcal{S}$ |
| $\underline{c}_s$ | default load of service $s \in \mathcal{S}$ (in TEU) |
| $\bar{c}_s$ | maximal load (capacity) of service $s \in \mathcal{S}$ (in TEU) |
| $\underline{m}_s$ | number of orders that make up for default load $\underline{c}_s$ of service $s \in \mathcal{S}$ |
| $r_s^{tport}$ | cost rate (\$/TEU) for transporting one TEU with service $s \in \mathcal{S}$ |
| $e_s^{tport}$ | emission rate ($kgCO_2e$/TEU) for transporting one TEU with service $s \in \mathcal{S}$ |
| decision variables | |
| $x_{os}$ | $= 1$ if order $o \in O$ uses service $s \in \mathcal{S}$, 0 otherwise |
| $y_{os}$ | $= 1$ if order $o \in O$ is transshipped at the end of service $s \in \mathcal{S}$ (i.e., at node $b_s \in \mathcal{N}$), 0 otherwise |
| $z_s$ | $= 1$ if the egalitarian and $= 0$ if the payload-based allocation scheme is used for service $s \in \mathcal{S}$ |
| $E_s^{\underline{c}}$ | emissions allocated to default orders $\underline{m}_s$ of service $s \in \mathcal{S}$ |
| $E_{os}^{tport}$ | emissions allocated to order $o \in O$ for using service $s \in \mathcal{S}$ |
| $E_{oi}^{tship}$ | emissions from transshipping order $o \in O$ at node $i \in \mathcal{N}$ |
| $E_o^{ex}$ | excess emissions, i.e., the amount of emissions allocated to order $o \in O$ that exceeds emission limit $\bar{e}_o$ |
| $E_o^{lim}$ | $= 1$ if emissions allocated to order $o$ exceed the order's emission limit $\bar{e}_o$ ($E_o^{ex} > 0$), 0 otherwise |

experiments, we consider orders each of a size of one or more container units (TEU) as these are candidates for intermodal transportation. However, the presented formulation of the SSP-TALE also fits to scenarios with smaller order sizes.

## 3. The SSP-TALE model

### 3.1. Problem description and notation

In the following, we provide a formal description of the considered problem. An overview of the used notation is provided in Table 2. We model the transportation network as a graph $\mathcal{G}(\mathcal{N}, \mathcal{S})$. The set of nodes $\mathcal{N}$ represents the origins and destinations of orders and services. The set of arcs $\mathcal{S}$ describes all services that operate between the nodes of the network. Each service $s \in \mathcal{S}$ connects an origin node $a_s$ and a destination node $b_s$. We allow parallel services between two nodes in order to model services of different types (e.g., truck or train) that connect the same pair of nodes. The problem of the SSP-TALE is to find an optimal routing plan for a given set of orders $O$ in network $\mathcal{G}$, as is conceptually sketched in Fig. 1.

For each order $o \in O$, a number $q_o$ of 20-ft container units (TEU) is to be shipped from a given origin node $a_o \in \mathcal{N}$ to a given destination node $b_o \in \mathcal{N}$ ($a_o \neq b_o$). In our study, each TEU corresponds to 12 tons of cargo, such that two TEU are equivalent to one full truck load of a 40-ft container à 24 tons.

The binary decision variable $x_{os}$ is used to denote if order $o$ uses service $s$ ($x_{os} = 1$), or not ($x_{os} = 0$). Similarly, binary decision variable $y_{os}$ is used to denote if order $o$ requires transshipment at the destination $b_s$ of service $s$ ($y_{os} = 1$), or not ($y_{os} = 0$). To identify transshipment processes, the parameter $suc_s \in \mathcal{S}$ indicates the succeeding service of a service $s \in \mathcal{S}$. For example, if two adjacent train services $s_1$ and $s_2$ with $b_{s_1} = a_{s_2}$ are conducted by the same train, we have $suc_{s_1} = s_2$. Then, an order that uses both services $s_1$ and $s_2$ requires no transshipment at location $b_{s_1}$ as it stays on the train. However, transshipment is required at node $b_{s_1} = a_{s_2}$ if $s_2$ is not conducted by the same train, which is indicated by parameter $suc_{s_1} \neq s_2$. Note that using binary decision variables to indicate if an order $o$ is transported with or transshipped after a service $s$ implies that payload $q_o$ is not split when being transported from $a_o$ to $b_o$. Transporting one TEU with service $s \in \mathcal{S}$ causes costs $r_s^{tport}$ and emissions $e_s^{tport}$. Transshipping one TEU at node $i \in \mathcal{N}$ causes costs $r_i^{tship}$ and emissions $e_i^{tship}$. For a specific service $s \in \mathcal{S}$, we denote by $\bar{c}_s$ the capacity and by $\underline{c}_s$ the default load of that service, both expressed in TEU. Here, default load $\underline{c}_s$ refers to the amount of transport units that use service $s$ independently of the routing of orders $O$. Thus, the total load of a service $s$ composes of its default load $\underline{c}_s$ and order load $\sum_{o\in O} x_{os} \cdot q_o$. The number of orders that make up for default load $\underline{c}_s$ is denoted by $\underline{m}_s$.

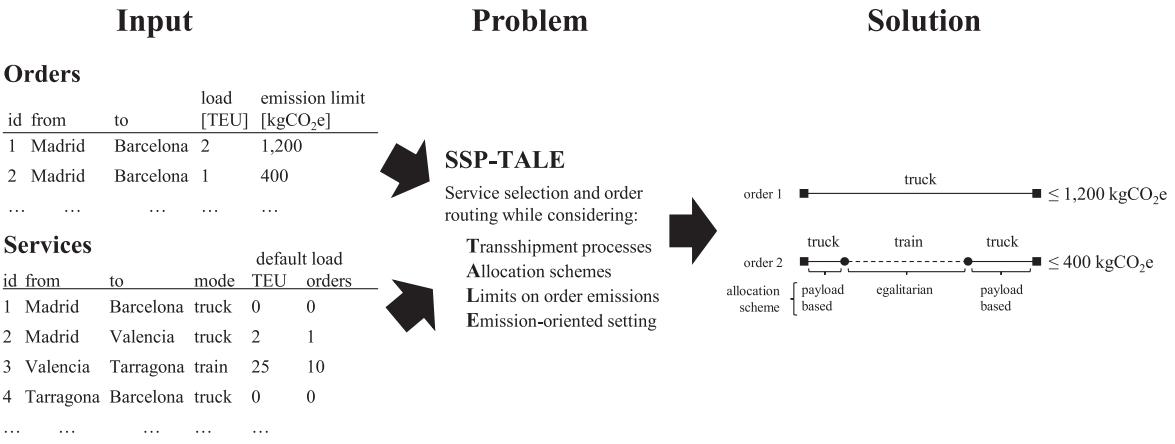



**Fig. 1.** Structure of the service selection problem with transshipment processes, allocation schemes per service and limits on order emissions in an emission-oriented setting (SSP-TALE).

We use equivalents of carbon dioxide ($CO_2e$) to subsume the environmental impact of several relevant greenhouse gases and refer to these jointly as "emissions" (e.g., Piecyk et al., 2012). The SSP-TALE considers emission limits set by customers. Such a limit $\bar{e}_o$ states the maximal amount of emissions (including emissions from transportation as well as emissions from transshipment) that is allowed for order $o \in O$. We use five decision variables to capture emissions: (i) $E_s^c$ states the amount of emissions of a service $s$ that are allocated to its default orders $\underline{m}_s$, (ii) $E_{os}^{tport}$ states the transport emissions of a service $s$ that are allocated to order $o$, (iii) $E_{oi}^{tship}$ states the transshipment emissions of order $o$ at a node $i \in \mathcal{N}$, (iv) $E_o^{ex}$ states the amount of emissions that exceeds the emission limit $\bar{e}_o$ of order $o$ and (v) $E_o^{lim}$ is a binary variable to indicate if emissions of order $o$ are higher than its limit ($E_o^{ex} > 0 \Rightarrow E_o^{lim} = 1$), or not ($E_o^{ex} = 0 \Rightarrow E_o^{lim} = 0$). Furthermore, the binary decision variable $z_s$ is equal to 1 if the egalitarian scheme (EA) is used for allocating emissions to orders of service $s$. If $z_s = 0$, the payload-based allocation scheme (PA) is used for this service.

For a service $s \in \mathcal{S}$, the egalitarian scheme ($z_s = 1$) allocates the same amount of emissions to all orders that use the service, independent of each order's actual payload. Thereby, the set of orders comprises both, the $\underline{m}_s$ default load orders and those orders $o \in O$ that are assigned to the service by the model (i.e., $x_{os} = 1$). For an order $o \in O$ that uses service $s \in \mathcal{S}$, the allocated transport emissions $E_{os}^{tport}$ are then determined by Formula (1). Here, the nominator describes the total transport emissions $e_s^{tport} \cdot (\underline{c}_s + \sum_{o' \in O} q_{o'} \cdot x_{o's})$ of service $s$ and the denominator corresponds to the total number of orders $\underline{m}_s + \sum_{o' \in O} x_{o's}$ that use service $s$. Note that Formula (1) is non-linear. We provide in Section 3.2 a linearized formulation of the respective model constraints.

$$(z_s = 1 \wedge x_{os} = 1) \Rightarrow E_{os}^{tport} = \frac{e_s^{tport} \cdot \left( \underline{c}_s + \sum_{o' \in O} q_{o'} \cdot x_{o's} \right)}{\underline{m}_s + \sum_{o' \in O} x_{o's}}, \forall\, o \in O, s \in \mathcal{S} \tag{1}$$

In contrast, a service $s \in \mathcal{S}$ that uses the payload-based scheme ($z_s = 0$) allocates the same amount of emissions to each transported TEU and, thus, considers an order's contribution $q_o$ to the total load ($\underline{c}_s + \sum_{o' \in O} q_{o'} \cdot x_{o's}$) of this service, see Formula (2). Here, the total emissions of service $s$ are allocated equally among all TEUs of this service and are then multiplied with the number of TEUs that correspond to order $o$.

$$(z_s = 0 \wedge x_{os} = 1) \Rightarrow E_{os}^{tport} = q_o \cdot \frac{e_s^{tport} \cdot \left( \underline{c}_s + \sum_{o' \in O} q_{o'} \cdot x_{o's} \right)}{\underline{c}_s + \sum_{o' \in O} q_{o'} \cdot x_{o's}} = q_o \cdot e_s^{tport}, \forall\, o \in O, s \in \mathcal{S} \tag{2}$$

Note that, irrespective of the applied allocation scheme, transport emissions of service $s \in \mathcal{S}$ are only allocated to order $o \in O$ if the order uses this service, i.e., $(x_{os} = 0) \Rightarrow E_{os}^{tport} = 0$. Emissions from transshipping an order $o \in O$ at the end-node $b_s$ of service $s \in \mathcal{S}$ are fully assigned to the order, i.e., $E_{ob_s}^{tship} = q_o \cdot e_s^{tship} \cdot y_{ob_s}$. An illustrative example of the SSP-TALE model is provided in Appendix A.

### 3.2. Optimization model

In this section, we present a MIP formulation for the SSP-TALE, considering four performance measures F1 to F4 that can serve as objectives, see Functions (3a) to (3d). The orders' total costs and emissions from transportation and transshipment are calculated by performance measures "order costs" (F1) and "order emissions" (F2), respectively. The total excess emissions that result from exceeding orders' emission limits are calculated by performance measure "excess emissions" (F3). The number of orders that exceed their emission limit are calculated by performance measure "red orders" (F4). We solve this multi-objective problem hierarchically,

i.e., we consider one of these performance measures as primary objective and the others as subordinate objectives. For example, using F1 (order costs) as primary objective and F2 (order emissions) as secondary objective results in minimal emissions for order set $O$ within the cost-minimizing solution. Of course, such a multi-objective problem can also be solved with other techniques. Linear weighting might be used but it is then the task to find proper weights for performance measures of different units. We refer to Marler and Arora (2004), Jozefowiez et al. (2008), and Deb (2014) for an overview of multi-objective optimization. In our study, we use the hierarchical approach as it clearly distincts the roles of the performances measures. In our experiments in Section 4, we use each combination of two out of the four measures F1 to F4 as primary and subordinate objectives, which leads to a total of 12 hierarchical objective-pairs being considered.

$$\min \text{F1} = \sum_{o\in O}\sum_{s\in S} q_o\cdot(r_s^{tport}\cdot x_{os} + r_{b_s}^{tship}\cdot y_{os}) \quad \text{“order costs”} \tag{3a}$$

$$\min \text{F2} = \sum_{o\in O}\left(\sum_{s\in S} E_{os}^{tport} + \sum_{i\in N} E_{oi}^{tship}\right) \quad \text{“order emissions”} \tag{3b}$$

$$\min \text{F3} = \sum_{o\in O} E_o^{ex} \quad \text{“excess emissions”} \tag{3c}$$

$$\min \text{F4} = \sum_{o\in O} E_o^{lim} \quad \text{“red orders”} \tag{3d}$$

subject to

$$\sum_{s\in S|a_s=a_o} x_{os} = 1,\ \forall\ o\in O \tag{4}$$

$$\sum_{s\in S|b_s=b_o} x_{os} = 1,\ \forall\ o\in O \tag{5}$$

$$\sum_{s\in S|b_s=i} x_{os} = \sum_{s\in S|a_s=i} x_{os},\ \forall\ o\in O,\ i\in N\backslash\{a_o, b_o\} \tag{6}$$

$$\sum_{s\in S|b_s=a_o} x_{os} + \sum_{s\in S|a_s=b_o} x_{os} = 0,\ \forall\ o\in O \tag{7}$$

Constraint (4) states that each order $o\in O$ has to depart at its origin $a_o$ and Constraint (5) states that the order has to arrive at its destination $b_o$. Constraint (6) describes each order's balance of flow at all other nodes of the network, i.e., an order which enters a node also has to leave that node. Constraint (7) prevents subtours by prohibiting orders to use services that arrive at their origin $a_o$ or depart at their destination $b_o$. Thus, the constraint ensures for all orders a cycle-free routing from the origin- to the destination-nodes.

$$y_{os} \geqslant x_{os} - x_{o,suc_s},\ \forall\ o\in O,\ s\in S \tag{8}$$

$$\underline{c}_s + \sum_{o\in O} q_o\cdot x_{os} \leqslant \bar{c}_s,\ \forall\ s\in S \tag{9}$$

Constraint (8) is used to set the transshipment decision variable $y_{os}$. Here, a transshipment operation $y_{os}$ for an order $o$ after using a service $s$ is only required if service's $s$ succeeding service $suc_s$ is not used by this order. Decision variable $y_{os}$ is never set unintentionally to value 1 as transshipment operations result in costs and emissions, one of which is always minimized in objectives F1 to F4. Constraint (9) describes the capacity restrictions of the services. Note that this constraint also implies that the complete order quantity $q_o$ is shipped with service $s$ if this service is used by order $o$ ($x_{os}=1$).

$$E_{oi}^{tship} = e_i^{tship}\cdot q_o\cdot \sum_{s\in S|b_s=i} y_{os},\ \forall\ o\in O,\ i\in N \tag{10}$$

$$E_s^c + \sum_{o\in O} E_{os}^{tport} = e_s^{tport}\cdot\left(\underline{c}_s + \sum_{o\in O} q_o\cdot x_{os}\right),\ \forall\ s\in S \tag{11}$$

$$E_{os}^{tport} \leqslant M_{os}^1\cdot x_{os},\ \forall\ o\in O,\ s\in S \tag{12}$$

$$E_s^c = 0,\ \forall\ s\in S|\underline{m}_s = 0 \tag{13}$$

Emissions that result from transshipment operations are calculated by Constraint (10) and emissions that result from transportation are calculated by Constraint (11). Note that Constraint (11) also ensures that all transport emissions are distributed among the default load and the orders $o\in O$, which implies that all emissions of a service are allocated. Constraint (12) ensures that emissions from a service $s$ can only be allocated to an order $o$ if this order uses service $s$ ($x_{os}=1$). Here, $M_{os}^1$ is a sufficiently large positive number ("big $M$") whose value is defined in Appendix B together with all other $M$-values appearing in the following constraints. Constraint (13) states that no emissions $E_s^c$ can be allocated to a service if a service has no default load ($\underline{m}_s = 0$).

$$E_{o_1 s}^{tport} - E_{o_2 s}^{tport} \leqslant M_{o_1 s}^{2} \cdot (3 - x_{o_1 s} - x_{o_2 s} - z_s),\ \forall\ o_1, o_2 \in O, s \in \mathcal{S} | \underline{m}_s = 0 \tag{14}$$

$$E_{o_2 s}^{tport} - E_{o_1 s}^{tport} \leqslant M_{o_2 s}^{2} \cdot (3 - x_{o_1 s} - x_{o_2 s} - z_s),\ \forall\ o_1, o_2 \in O, s \in \mathcal{S} | \underline{m}_s = 0 \tag{15}$$

$$\frac{E_s^{\underline{c}}}{\underline{m}_s} - E_{os}^{tport} \leqslant M_s^{3} \cdot (2 - x_{os} - z_s),\ \forall\ o \in O, s \in \mathcal{S} | \underline{m}_s > 0 \tag{16}$$

$$E_{os}^{tport} - \frac{E_s^{\underline{c}}}{\underline{m}_s} \leqslant M_{os}^{4} \cdot (2 - x_{os} - z_s),\ \forall\ o \in O, s \in \mathcal{S} | \underline{m}_s > 0 \tag{17}$$

The egalitarian allocation of emissions to orders in $O$ is described by Constraints (14)–(17). Services with no default load ($\underline{m}_s = 0$) are treated in (14) and (15). They allocate the same emissions pairwise to orders $o_1$ and $o_2$ that both use service $s$. Constraints (16) and (17) handle services that have a default load ($\underline{m}_s > 0$). Here, the emissions $E_{os}^{tport}$ allocated to order $o$ are set equal to the emissions allocated to each order of the default load which is computed by $\frac{E_s^{\underline{c}}}{\underline{m}_s}$. Constraints (14) and (15) are based on the implementation of the egalitarian scheme in the selective traveling salesman problem of Kirschstein and Bierwirth (2018). This implementation is a linearization of the generic but non-linear Eq. (1) that was presented in Section 3.1.

$$E_s^{\underline{c}} - e_s^{tport} \cdot \underline{c}_s \leqslant M_s^{5} \cdot z_s,\ \forall\ s \in \mathcal{S} \tag{18}$$

$$e_s^{tport} \cdot \underline{c}_s - E_s^{\underline{c}} \leqslant M_s^{6} \cdot z_s,\ \forall\ s \in \mathcal{S} \tag{19}$$

$$E_{os}^{tport} - e_s^{tport} \cdot q_o \cdot x_{os} \leqslant M_{os}^{7} \cdot (1 - x_{os} + z_s),\ \forall\ o \in O, s \in \mathcal{S} \tag{20}$$

$$e_s^{tport} \cdot q_o \cdot x_{os} - E_{os}^{tport} \leqslant M_{os}^{8} \cdot (1 - x_{os} + z_s),\ \forall\ o \in O, s \in \mathcal{S} \tag{21}$$

The payload-based allocation of emissions is modeled by Constraints (18) and (19) for the default load of service $s$ and by Constraints (20) and (21) for the transport emissions $E_{os}^{tport}$ of order $o$ using service $s$. Remember that $z_s = 0$ indicates that the PA-scheme is applied to service $s$.

$$E_o^{ex} \geqslant \sum_{s \in \mathcal{S}} E_{os}^{tport} + \sum_{i \in N} E_{oi}^{tship} - \bar{e}_o,\ \forall\ o \in O \tag{22}$$

$$E_o^{ex} \leqslant M_o^{9} \cdot E_o^{lim},\ \forall\ o \in O \tag{23}$$

$$E_s^{\underline{c}},\ E_{os}^{tport},\ E_{oi}^{tship},\ E_o^{ex} \geqslant 0,\ \forall\ o \in O, s \in \mathcal{S}, i \in \mathcal{N} \tag{24}$$

$$x_{os},\ y_{os},\ z_s,\ E_o^{lim} \in \{0, 1\},\ \forall\ o \in O, s \in \mathcal{S} \tag{25}$$

Constraint (22) calculates the emissions that exceed order $o$'s emission limit $\bar{e}_o$. If this value is strictly positive, the binary indicator $E_o^{lim}$ is set to 1 by Constraint (23). Note that $E_o^{ex}$ and $E_o^{lim}$ are never unintentionally larger than 0 if these measures are considered as primary or subordinate objectives. Finally, the domains of the decision variables are described by Constraints (24) and (25). The implementation is enhanced by bounds that limit the range of possible values for decision variables $E_s^{\underline{c}}$, $E_{oi}^{tship}$, and $E_{os}^{tport}$. A description of these bounds is provided together with the large numbers $M$ in Appendix B. We use version 12.7.1.0 of IBM ILOG CPLEX for solving this formulation of the SSP-TALE.

## 4. Experiments with real-world data

### 4.1. Data

For the following experiments, we use the SSP-TALE model to route a fixed number of orders through the European intermodal rail/road network. Thereby, we compare results from country pairs Portugal and the Netherlands (PT-NL) as well as Italy and Denmark (IT-DK). Orders between these countries can be candidates for intermodal rail/road transportation due to the long distances. The results from these country pairs are used to demonstrate the impact of SSP-TALE's features whereas for other country pairs results might differ. For example, Arnold et al. (2004) show that modal shares are also sensitive to costs of track gauge changes which could be incorporated when considering countries with differing track gauge.

The considered rail network is based on the core corridors described in the Trans-European Transport Network (TEN-T), see PaP Catalogue (2019) and TENtec Reporting (2014, 2019). Fig. 2a illustrates this network. Full data of the network is provided in Appendix C. Road network data is extracted from Open Street Map (2019) using Geofabrik (2019). We apply the Open Source Routing Machine to obtain truck routes within this road network, see Luxen and Vetter (2011). Fig. 2b shows an exemplary routing through the road network from a customer node in Portugal to five rail/road transshipment terminals on the Iberian Peninsula.

For the rail services, we model an artificial mix of light and regular trains, see Fig. 2a. Here, regular trains operate on main axes across Europe, such as the "west-to-east" axis or "north-to-south" axis, and light trains are used as feeders to the regular trains. Overall, we model 88 light trains and 18 regular trains that are also described in Appendix C. We assume for both train types a maximum capacity of $\bar{c}_s = 108$ TEUs, which corresponds to 36 wagons à 3 TEU. For the default load of a service $s$, we use values of $\underline{c}_s = 25$ TEU for light trains and $\underline{c}_s = 40$ TEU for regular trains. These default loads are chosen because they pose constellations where

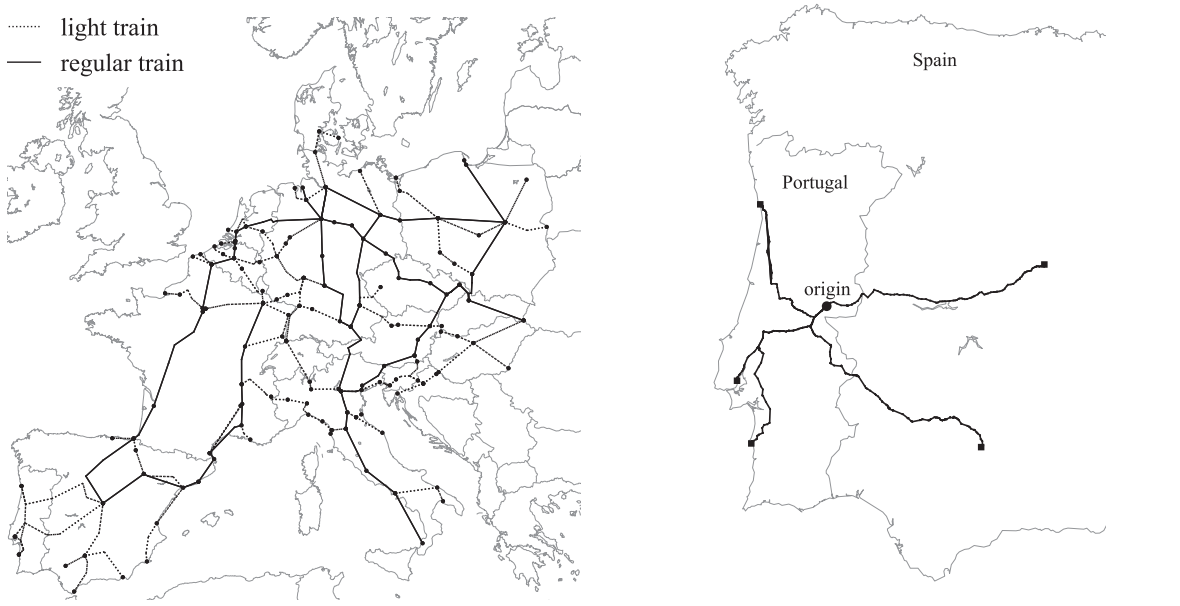



**Fig. 2.** Considered rail network and exemplary routing through the road network.

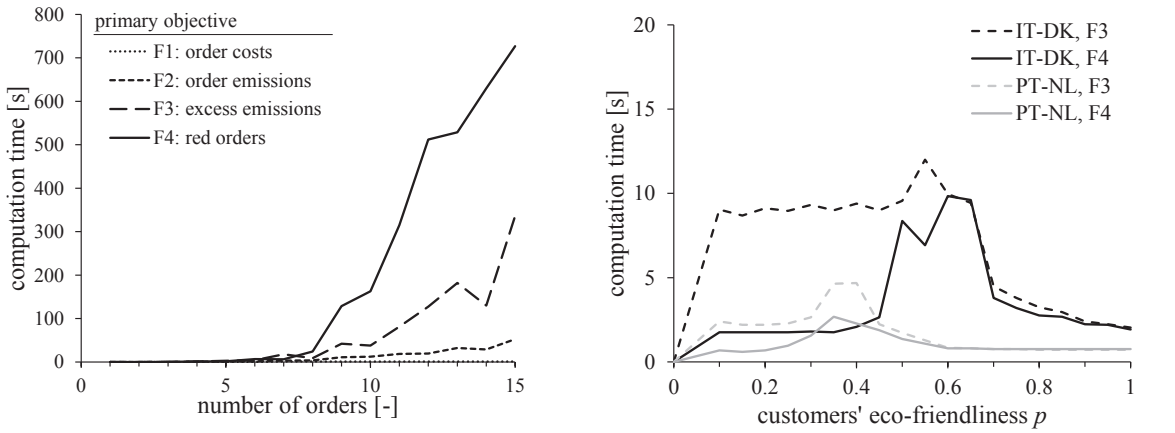
the selection of the services' allocation schemes is not trivial. This is further explained in Section 4.6, where we present results for varying values of $\underline{c}_s$. A train's remaining capacity $\bar{c}_s - \underline{c}_s$ can be used for loads $q_o$ of orders $O$. The number of external orders $\underline{m}_s$ is randomly drawn between 1 and 10 for both train types. For the road services, we model heavy trucks. We consider road services that connect each order's origin location directly to the five closest transshipment terminals. Likewise, road services that connect a destination location to five nearby terminals are established. For an order $o$, and each such road service $s$, we assume that the maximum capacity of the road service equals the quantity of the related order ($\bar{c}_s = q_o$). Thus, as each truck can carry at most two TEU, a road service can correspond to multiple trucks. Road services $s$ are assumed to have no default load ($\underline{c}_s = \underline{m}_s = 0$).

Throughout this study, we assume transshipment costs of $r_i^{tship} = 40$ \$/TEU and transshipment emissions of $e_i^{tship} = 15$ $kgCO_2e$/TEU for all nodes $i \in N$, see Clausen et al. (2013), Geerlings and van Duin (2011), and Janic (2007). The services' cost rates $r_s^{tport}$ (\$/TEU) are 1.67 \$ cents/ton·km for train transportation and 5.23 \$ cents/ton·km for truck transportation (Forkenbrock, 2001). The services' emission rates $e_s^{tport}$ ($kgCO_2e$/TEU) of truck and train freight transportation are calculated with average country-pair specific values provided by Heinold and Meisel (2018), see Table 3. Emission rates differ between country pairs because countries have their own geographical and/or infrastructural characteristic (e.g., crossing the Alps, density of the rail network, degree of electrification in the rail network, source of electricity, etc.) and they differ between light and regular trains because each train type has a different default weight resulting from default load $\underline{c}_s$.

For each order $o \in O$, a random origin $a_o$ and a random destination $b_o$ are drawn within the borders of the respective country. We draw order quantities $q_o$ between 1 and 8 TEUs. For the emission limit $\bar{e}_o$ of an order $o$, we multiply the order's road-only emissions with a weight $p$. Here, road-only emissions of an order $o$ refer to the emissions that result from shipping the load $q_o$ directly by truck from the order's origin to its destination. Note that such truck services are not contained in set $S$ as we look for analyzing intermodal operations. For eco-aware customers, it is reasonable to assume that $p < 1$ as transporting goods solely by truck usually causes higher emissions than using intermodal rail/road transportation (e.g., Heinold and Meisel, 2018). In other words, parameter $p$ describes the share of road-only emissions that a customer is willing to accept for an intermodal transportation. However, it is difficult to preset a particular value of $p$ as it is an indicator for the environmental cautiousness of an individual customer. Therefore, we vary this parameter in our experiment in Section 4.3.

**Table 3**
Emission rates from truck and train freight transportation.

| | emission rate [$gCO_2e$/ton·km] | | |
|---|---|---|---|
| country pair | light train | regular train | truck |
| PT-NL | 10.0 | 9.3 | 53.3 |
| IT-DK | 24.9 | 23.4 | 59.4 |




(a) different number of orders (PT-NL, $p = 0.35$). (b) varying values of parameter $p$ (5 orders).

**Fig. 3.** Average computation time for a varying number of orders and varying values of $p$.

All results presented in this paper are based on solving 25 instances for each of the considered country pairs, using either five or eight orders per instance ($|O| = \{5, 8\}$). Note that by considering at most 8 orders of 8 TEUs each, the capacities of trains are not a limiting factor here. This relatively small number of orders already represents a realistic setting for the considered problem. For example, if all of the 8 orders are transported with the same train service, these orders could make up for nearly 60% of this train's capacity. Furthermore, the services' default loads can be seen as several independent orders that have already been assigned to these services, for example, in previous planning periods. These orders are assumed here to have no preferences regarding their environmental impact. Thus, the default load of a service $s$ could also be modeled as many artificial orders between the service's origin $a_s$ and destination $b_s$. However, by modeling them collectively as "default load", it is possible to reduce the computational challenge of solving the problem. A more enhanced analysis on the computation time is provided in the following section.

### 4.2. Computation time

This first experiment analyzes the computation time that is required for solving the SSP-TALE instances. The experiment is conducted on a computer with 32 GB RAM and an i7-8700 CPU processor with 3.2 GHz. The computation time depends on several factors which are interdependent with each other. We start reporting computation times under objectives F1 to F4 for instances with a varying number of orders. Fig. 3a shows the average computation time (in seconds) for shipping 1 to 15 orders from Portugal to the Netherlands. As expected, more orders lead to a higher computation time. More interestingly, the computational challenge notably rises for objectives "excess emissions" (F3) and "red orders" (F4), which are the two novel objectives introduced in this study. This is because F3 and F4 consider whether an order is routed within its emission limit $\bar{e}_o$, which requires the model to take into account decision variables $E_o^{ex}$ and $E_o^{lim}$, respectively, and to make order-specific allocation decisions. In contrast, these variables can be ignored in the solution process under objectives F1 and F2. Furthermore, it is not even possible to find an optimal solution for every large instance under objectives F3 and F4 within a given runtime of 60 min. For example, this holds for 6 out of 25 instances with 15 orders under objective F4. These instances were therefore excluded from the results in Fig. 3. Note that the results are obtained from strengthened models that use the cuts and big-M values in Appendix B. For example, if instances 1–10 (5 orders, PT-NL, objective F4, flexible EA/PA allocation) are solved without these cuts, no optimal solution is found within 300 s for four instances (average gap 40%) and the average computation time of the remaining six instances is 79 s. In contrast, using the cuts, an optimal solution for all ten instances is found within an average runtime of 2 s.

Fig. 3b shows the average computation time under the more time-consuming objectives F3 and F4 for shipping five orders between PT-NL and IT-DK under varying values of the customers' eco-friendliness $p$. It can be seen that the computation time behaves differently for the two country pairs. This is because the country-specific emission rates and the different network structures offer diverse country-specific options for routing orders and for respecting emission limits. In particular, there exist more possible routings for orders from Italy to Denmark compared to orders from Portugal to the Netherlands. The figure also shows that it is relatively easy to solve instances where $p$ is either very low or very high. For extreme values of $p$, it is relatively easy for the model to find an optimal solution as either all or none of the orders can be shipped within their emission limit. Consequently, it is most challenging to come up with an optimal solution if customers ask for a medium level of eco-friendliness. A similar observation can be made for varying values of the services' default loads. Here, it is not challenging for the model to find an optimal solution if the selection of the services' allocation schemes is straightforward. For example, if the services' default load is relatively large compared to the size of the orders that are to be routed, the egalitarian scheme always allocates more emissions to the orders than the payload scheme and, thus, the payload scheme is used under the service-oriented objectives F3 and F4.

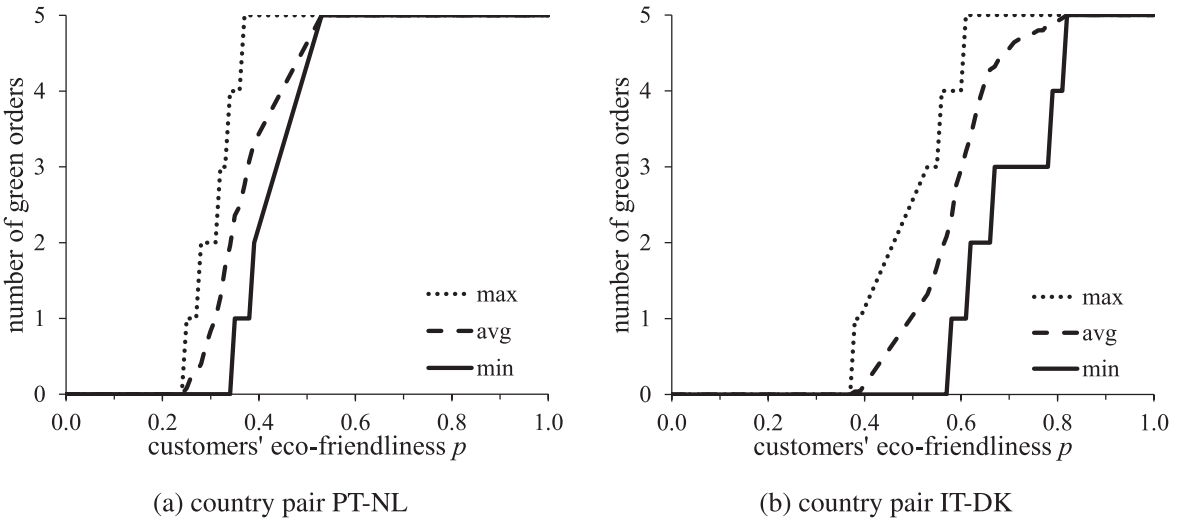



**Fig. 4.** Number of green orders for varying values of $p$.

Overall, these results show that the computation time depends on several factors. The most important ones are as follows: (i) considered number of orders $|O|$, (ii) chosen objective, (iii) considered country pair, (iv) level of customers' eco-friendliness $p$, and (v) difficulty to select the services' allocation schemes (depends on order size and default load).

### 4.3. Customers eco-friendliness (p-value)

We start by investigating the impact of parameter $p$, which indicates the level of eco-friendliness sought by the shippers of orders $O$. We analyze for varying values of $p$, the number of orders that do not exceed their emission limit (referred to as *green orders*), which is the difference between the total number of orders $|O|$ and the number of orders that exceed their emission limit (referred to as *red orders*). These results are obtained by using primary objective "red orders" (F4) which minimizes the number of red orders and, thus, maximizes the number of green orders.

Fig. 4 shows the minimal, average and maximal number of green orders over the 25 test instances for both country pairs with five orders per instance, using varied values of $p \in [0, 1]$. If parameter $p = 1$, the minimal number of green orders is 5 for both country pairs. This means, that there is a solution for all instances in which all orders are routed within their emission limits if these limits are sufficiently high. In contrast, if $p = 0.35$ for country pair PT-NL, the minimal number of green orders is 1, the average is 2.36 and the maximum is 4 over the 25 test instances. Thus, if customers request that the intermodal transport's emissions are at most 35% of the direct road-only transport's emissions ($p = 0.35$), the model finds solutions where on average around half of the considered orders are routed within their limit.

For country pair IT-DK, if $p = 0.55$, the average number of green orders is 1.68. In addition, this $p$-value can result in solutions where no orders are routed within their emission limits (minimal number of green orders is 0) or where at most 3 orders are routed within their emission limit (maximal number of green orders). Thus, despite of a higher $p$-value, the average number of green orders is lower for country pair IT-DK compared to PT-NL. This is because country-specific emission rates for truck and train transportation are used and their ratio is closely related to the impact of the $p$-value on the number of green orders. For example, the ratios of emissions from truck and regular trains are 0.39 for IT-DK but only 0.17 for PT-NL. In other words, train transportation between Portugal and the Netherlands causes relatively low emissions compared to Italy and Denmark, which allows, for the same $p$-value, more green orders for country pair PT-NL compared to IT-DK. Due to this country-pair specific spread, the further results presented in this paper use $p = 0.35$ for country pair PT-NL and $p = 0.55$ for country pair IT-DK as these represent interesting constellations for the considered problem.

### 4.4. Emission limits and order emissions

We analyze in this section the impact of primary objectives F1 to F4 on the emissions allocated to the orders. For this, we show exemplary results from a selected test instance with five orders between Portugal and the Netherlands, see Fig. 5. We use objective "order emissions" (F2) as secondary objective and the values atop of the bars measure the difference between the emissions allocated to the orders and their respective emission limit (shown as black bar in Fig. 5). This value is measured in $tCO_2e$ and positive (+) for the case of red orders and negative (-) for the case of green orders. There are three red orders (2, 4 and 5) under objective "order costs" (Fig. 5a), two red orders (4 and 5) under objectives "order emissions" and "excess emissions" (Figs. 5b and c), and only one red order (4) under objective "red orders" (Fig. 5d). Thus, the number of red orders differs among the solutions and reaches its minimum only under the corresponding objective "red orders" (F4). In contrast, objective F2 minimizes the total emissions that are allocated to

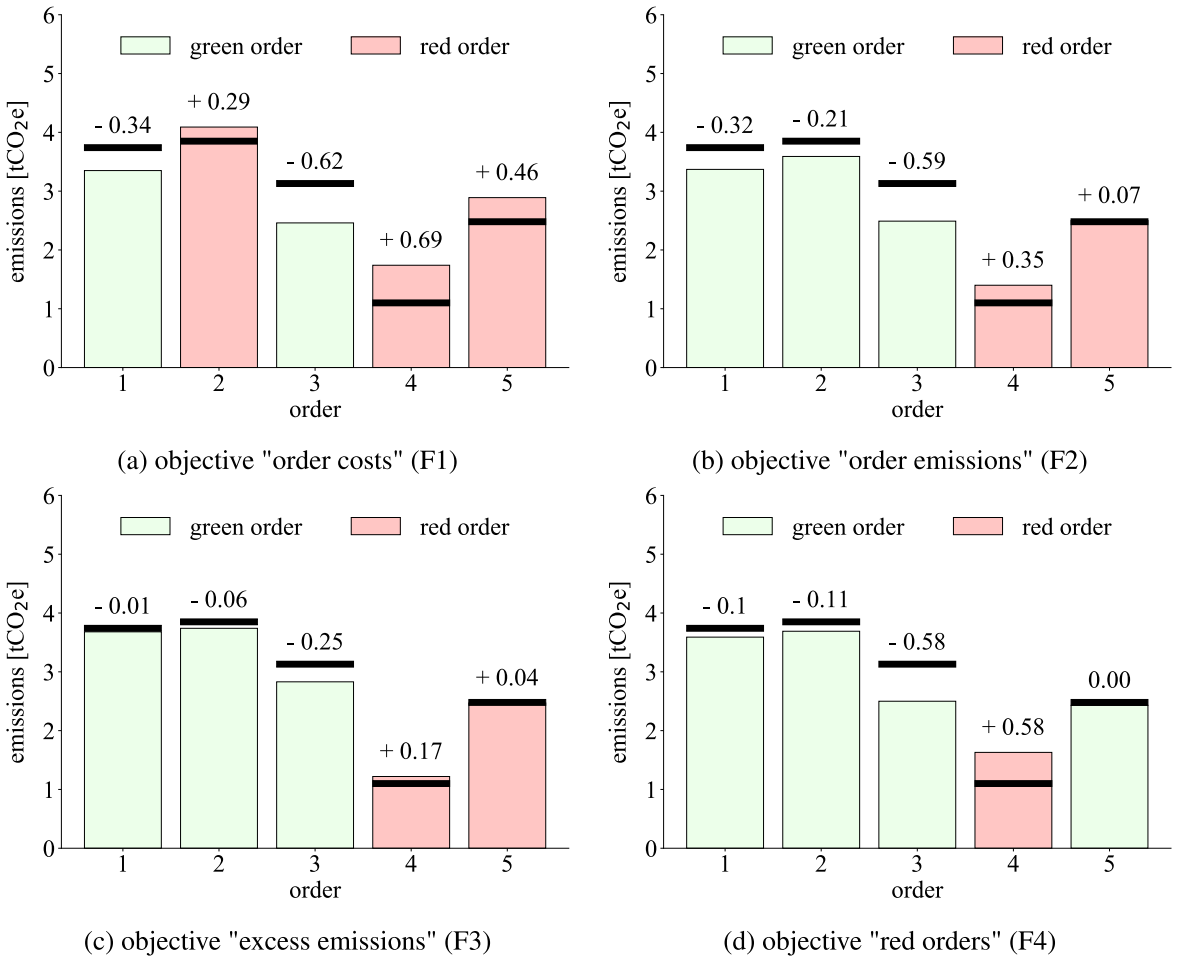



**Fig. 5.** Emissions per order under primary objectives F1 to F4 (instance 17, PT-NL).

the orders which, at the same time, implies two red orders (4 and 5). Consider now that the emission limit of order 5 increases by 0.07 $tCO_2e$ (which is the order's excess emission under F2). Then, the solution that minimizes the total order emissions (F2) and the solution that minimizes the total number of red orders (F4) would be the same. However, using the emission limits as they are, order 5 becomes "green" under objective F4 only by allocating more emissions to the other orders 1 to 4. This can be done by changing the services' allocation schemes or by changing the other orders' routings. Either way, this shows that the allocation of emissions to a particular order also depends on the emission limits of other orders. We refer to the second example in Appendix A for a further illustration of this finding.

Note that each solution in Fig. 5 comes up with a different allocation of emissions. In other words, total order emissions increase by considering the orders' emission limits. Fig. 6 further illustrates this by showing the emissions that are allocated to the orders (performance measure F2) in each of the 25 PT-NL instances with five orders, using F1 to F4 as primary objectives. In the figure, the orders are sorted by decreasing values of the order emissions under secondary objective F2. It can be seen that the emissions allocated to the orders under the different objectives constitute a lower bound and an upper bound. The lower bound is obtained for all instances if "order emissions" (F2) is considered as primary objective. The upper bound follows from primary objective "order costs" (F1) for 18 instances, from objective "excess emissions" (F3) for four instances and from objective "red orders" (F4) for three instances. Thus, as expected, using performance measures related to emission limits (F3 and F4), usually lead to solutions where less emissions are allocated to the considered orders than in minimum-cost solutions. At the same time, there is no substantial difference in the emissions allocated to the orders between objectives "excess emissions" (F3) and "red orders" (F4).

### *4.5. Allocation schemes and order size*

So far, the results are based on the assumption that the model selects the emission allocation scheme for each service individually. To analyze the impact of this degree of freedom, we now look at the case where all services exclusively use either the payload-based or the egalitarian allocation scheme. Fig. 7 shows the total emissions (F2) that are allocated to the orders for the 25 instances of country pair PT-NL (five orders each), considering the three allocation policies (exclusively EA; exclusively PA; flexible EA/PA). Here,

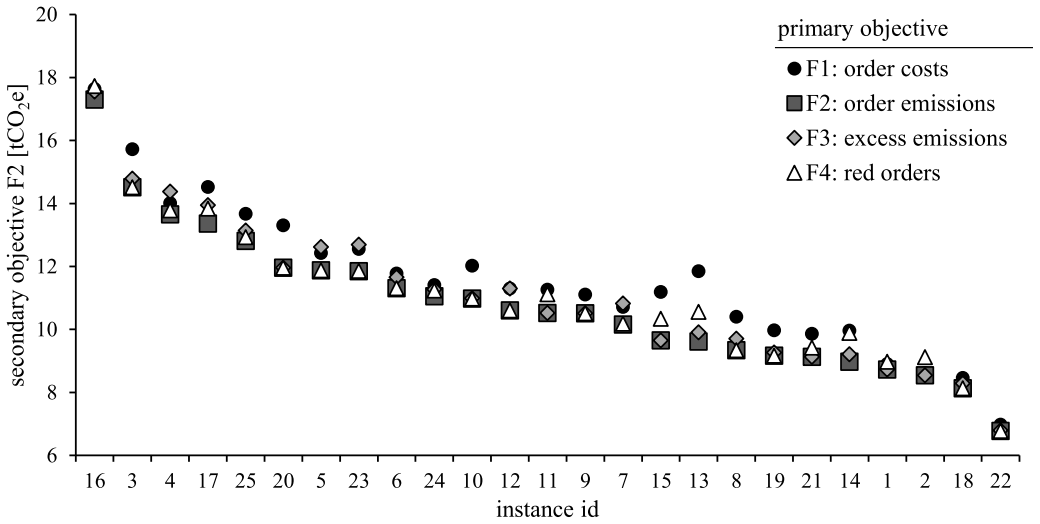



**Fig. 6.** Emissions per instance under primary objectives F1 to F4 (five orders, PT-NL).

the instances are sorted by their total order size ($\sum_{o \in O} q_o$), where 13 TEU and 35 TEU is the lowest and highest total order size, respectively. We make the following observations. If instances have a small total order size, e.g., below about 20 TEU, the exclusive PA scheme allocates less emissions to the considered orders than the exclusive EA scheme. In contrast, if instances have a higher order size, the exclusive EA scheme allocates less emissions to the considered orders than the exclusive PA scheme. Thus, the size of the instance appears to be a good indicator for comparing the emissions that are allocated under the two exclusive policies. Moreover, this observation for five orders per instance seems to be very similar to the allocation scheme that is selected if a single order uses a service. Here, if the order is relatively small, then lowest emissions are allocated to the order under the payload-based scheme. Likewise, if the order is relatively large, then lowest emissions are allocated to the order under the egalitarian scheme. This is because the egalitarian scheme implicitly balances the emissions caused by the default load with the emissions caused by the order, which is more beneficial if the considered order is relatively large. Section 4.6 provides additional insights on this topic by presenting results for varying values of the services' default loads. We refer to Appendix A for an illustrative example with a single order. Anyhow, for the instances considered in this study, it can be seen that having the freedom of choosing the allocation scheme (flexible EA/PA) allocates lowest emissions to orders for all instances, which in most cases outperforms both of the pure allocation schemes. In general, the exclusive use of the PA (EA) scheme can only be optimal for instances with a relatively low (high) total order size. A mix of these two schemes is optimal for instances with a total order size that is in-between these two extremes.

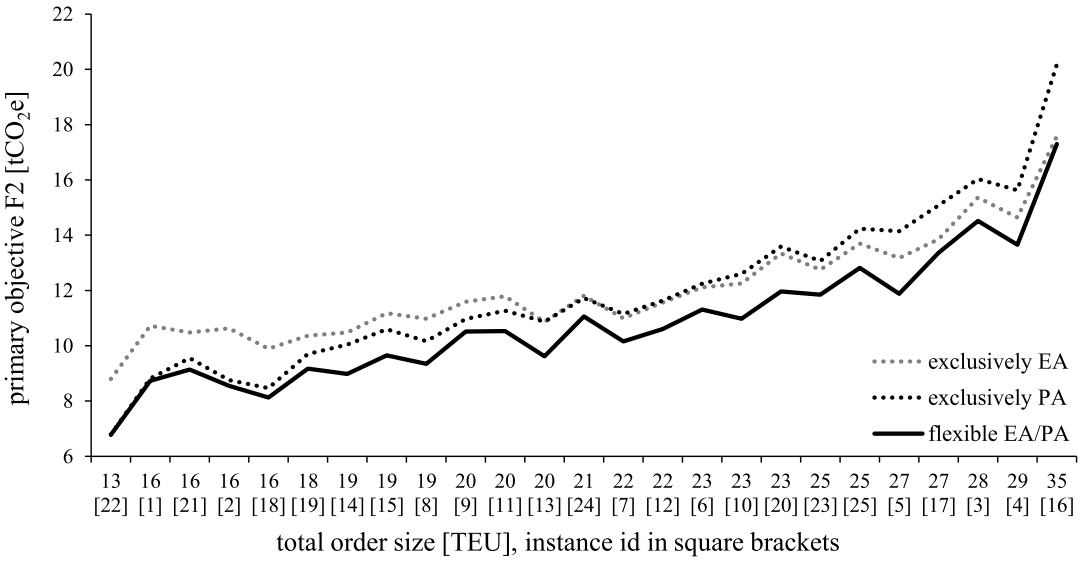



**Fig. 7.** Emissions per instance for three emission allocation cases (five orders, PT-NL).

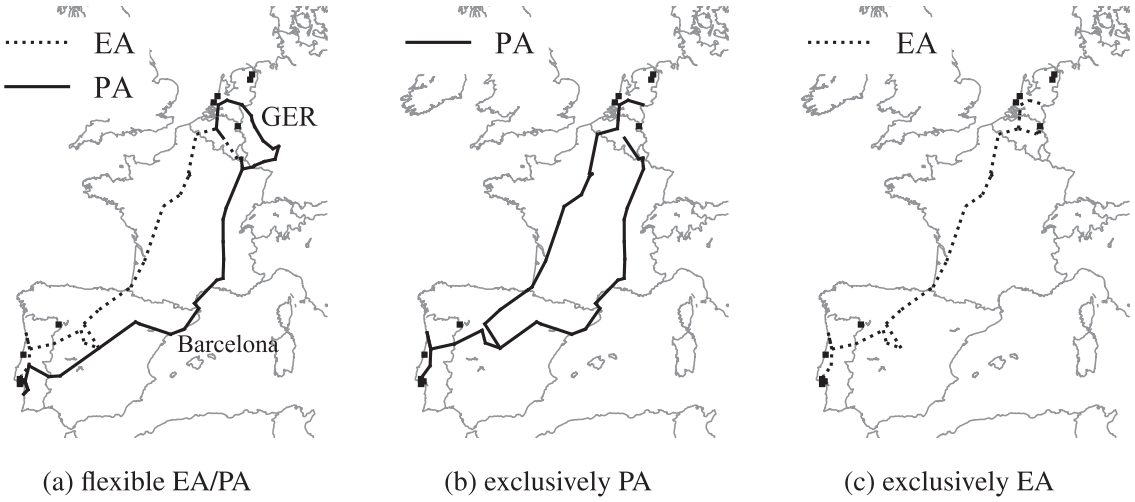



**Fig. 8.** Used rail services for three emission allocation policies (instance 10, five orders, PT-NL).

### *4.6. Allocation schemes and default load*

We now analyze the impact of the allocation schemes on the routing as well as the impact of the default load on the chosen allocation scheme. Fig. 8 demonstrates the impact on the orders' routing if services are forced to use a specific emission allocation scheme. The figure shows for a selected test instance the used services and their allocation schemes for each of the three allocation policies. It can be seen that the used services strongly differ for the three policies. For example, the services through Germany are only used if the model is free to select between the EA and PA allocation scheme (Fig. 8a) whereas the train services along Barcelona are only used if the PA scheme is allowed for these services (Fig. 8a and b). In this example, if the EA scheme is prescribed for all services, most orders use the same set of rail services (Fig. 8c).

The selection of the allocation scheme and, thus, the routing of the orders is strongly affected by the composition of the default load (size and number of orders). So far, the results are based on the assumption of having rail services with a fixed default load size $\underline{c}_s$ and a randomly drawn number of external orders $\underline{m}_s \in [1, 10]$. In the following experiment, we set the number of default orders to $\underline{m}_s = 5$ and solve the instances with default loads of $\underline{c}_s = 1, 2, \ldots, 50$ TEU. Fig. 9 shows the results under primary objective "order emissions" (F2) for the three emission allocation policies. We make the following observations. If the EA scheme is used exclusively, an increasing default load increases the emissions that are allocated to the orders. In contrast, these emissions are not affected if the PA scheme is used exclusively. The latter is because the emission rate is independent of the default load in this experiment. Furthermore, if the size of the services' default load $\underline{c}_s$ is low, then the EA scheme allocates less emissions to the considered orders

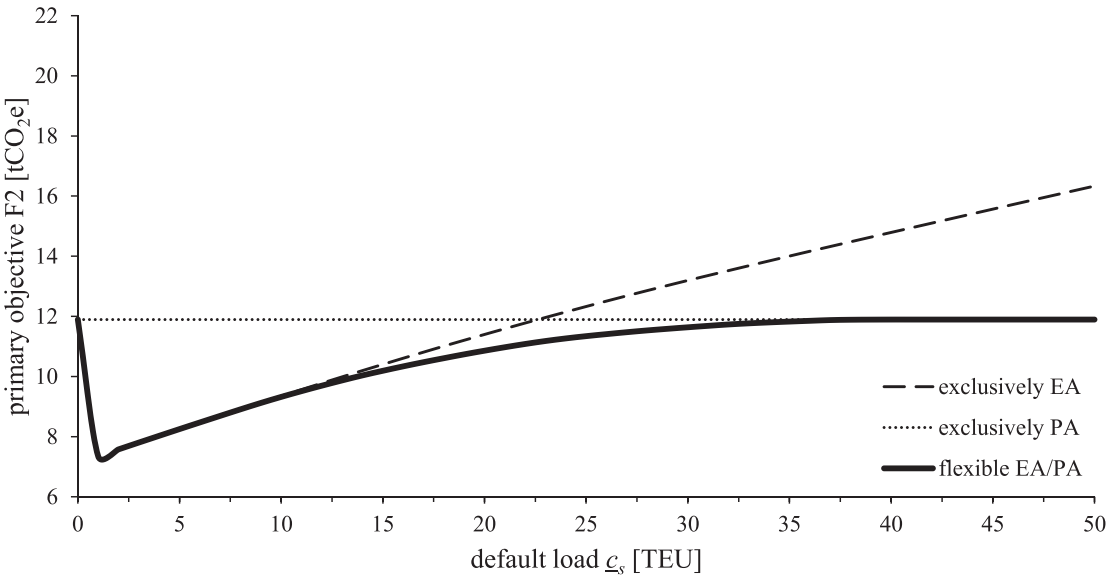



**Fig. 9.** Average emissions over all instances for varying values of default load (five orders, PT-NL).

**Table 4**
Average results from a hierarchical optimization over 25 instances.

| | | secondary objective | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | five orders | | | | eight orders | | | |
| | | F1 [thous. $] | F2 $tCO_2e$ | F3 $tCO_2e$ | F4 [red orders] | F1 [thous. $] | F2 $tCO_2e$ | F3 $tCO_2e$ | F4 [red orders] |
| primary objective | PT-NL ($p$ = 0.35) | | | | | | | | |
| | F1: order costs | **20.13**$^1$ | 11.59$^4$ | 1.38$^4$ | 3.36$^3$ | **33.61**$^1$ | 20.30$^4$ | 3.25$^4$ | 6.44$^2$ |
| | F2: order emissions | 20.68$^3$ | **10.82**$^1$ | 0.86$^2$ | 3.32$^2$ | 34.83$^4$ | **19.08**$^1$ | 2.25$^2$ | 6.52$^3$ |
| | F3: excess emissions | 20.72$^4$ | 11.10$^3$ | **0.79**$^1$ | 3.40$^4$ | 34.76$^3$ | 19.29$^2$ | **2.19**$^1$ | 6.52$^3$ |
| | F4: red orders | 20.26$^2$ | 11.05$^2$ | 0.97$^3$ | **2.64**$^1$ | 34.45$^2$ | 20.17$^3$ | 3.24$^3$ | **5.56**$^1$ |
| | IT-DK ($p$ = 0.55) | | | | | | | | |
| | F1: order costs | **15.66**$^1$ | 15.04$^3$ | 1.24$^3$ | 3.76$^2$ | **24.35**$^1$ | 23.24$^3$ | 2.05$^3$ | 5.68$^2$ |
| | F2: order emissions | 16.01$^4$ | **14.71**$^1$ | 1.03$^2$ | 3.88$^3$ | 24.88$^3$ | **22.39**$^1$ | 1.67$^2$ | 5.80$^3$ |
| | F3: excess emissions | 15.99$^3$ | 14.79$^2$ | **0.99**$^1$ | 4.04$^4$ | 24.90$^4$ | 22.89$^2$ | **1.45**$^1$ | 6.08$^4$ |
| | F4: red orders | 15.80$^2$ | 15.08$^4$ | 1.37$^4$ | **3.32**$^1$ | 24.66$^2$ | 23.46$^4$ | 2.54$^4$ | **4.52**$^1$ |

superscripts 1–4 refer to the rank of the column's measure (1 = lowest value; 4 = highest value)

compared to the PA scheme. Likewise, if the size of the services' default load $\underline{c}_s$ is high, then the PA scheme allocates less emissions to the considered orders compared to the EA scheme. However, if $\underline{c}_s$ is somewhere in-between, then the policy that flexibly combines both allocation schemes outperforms all pure allocation policies. Note that this observation is in line with the findings about the order size from Section 4.5. In fact, it is the relation between the order size and the default load size that jointly decides upon the allocation schemes that are used for the services.

### 4.7. Overall results for five and eight orders and both country pairs

The previous sections primarily presented results for five orders and country pair PT-NL. We now extend this analysis by considering five or eight orders that have to be shipped from Portugal to the Netherlands or from Italy to Denmark. Table 4 shows the results from a hierarchical optimization, using objectives F1 to F4 either as a primary objective and/or as a secondary objective. Results are presented as averages over the 25 instances for each country pair with either five or eight orders per instance. The values in the table correspond to the performance measures of the secondary objective, which is stated at the top of a column. For example, using F1 as primary objective gives minimal average values of F1 = 20.13 thous. $, F2 = 11.59 $tCO_2e$, F3 = 1.38 $tCO_2e$ and F4 = 3.36 red orders, see first line in Table 4. Conversely, from looking at the first column in Table 4, the average minimal costs that are allocated to the orders (F1) for shipping five orders from Portugal to the Netherlands are 20.13 under primary objective F1, 20.68 under F2, 20.72 under F3, and 20.26 under F4. The superscripts shown in the table refer to the rank of the secondary objective's value, where 1 refers to the lowest and 4 to the highest value. In addition, the bold values highlight the value of rank 1, as this is the minimum of the corresponding primary objective.

Using the ranks, it is easy to compare results for the secondary objective's performance measure. For example, the highest average number of red orders (F4) is always achieved under primary objective "excess emissions" (F3). In particular, the average number of red orders is 3.40 out of five orders and 6.52 out of eight orders for country pair PT-NL as well as 4.04 out of five orders and 6.08 out of eight orders for country pair IT-DK. However, the average difference in the number of red orders under primary objectives F1 to F3 is quite small for instances from both country pairs. For performance measure "excess emissions" (F3), we see a relatively high difference between the lowest and the highest value, which indicates that excess emissions are strongly influenced by the chosen primary objective. For example, this difference is 0.38 $tCO_2e$ (+39%) and 1.09 $tCO_2e$ (+76%) for five and eight orders shipped from Italy to Denmark. From this, it can be concluded that F3 chooses more orders where emissions exceed the limit by only a small amount and F4 chooses then a lower number of orders where the emissions exceed the limit by a somewhat higher amount per order. In contrast, the primary objective's relatives impact on performance measures F1 and F2 is less strong and lies between 2.2% and 3.6% for costs (F1) and between 2.5% and 7.1% for emissions (F2). In other words, using F2, F3, or F4 as primary objective increases costs by at most 3.6% and using F1, F3, or F4 as primary objective increases emissions by at most 7.1%. Overall, the results from Table 4 show the trade-offs that are implied by considering orders' emission limits. The basic findings from the previous sections apply to five and eight orders as well as to both country pairs. Neither minimizing total costs nor minimizing total emissions leads to a routing where as many orders as possible are routed within their emission limit, i.e., where the number of green orders is maximized.

### 4.8. Network results

Remember that the performance measures F1 to F4 solely capture the effects for order set $O$. Therefore, in this section, we provide results for measures that consider network-wide costs and emissions from both, the orders and the default load of services. In particular, we use performance measures "network costs" (F5) and "network emissions" (F6), see Formulas (26a) and (26b). Note that objectives "order costs" (F1) and "network costs" (F5) always lead to the same routing. This is because we consider no cost allocation

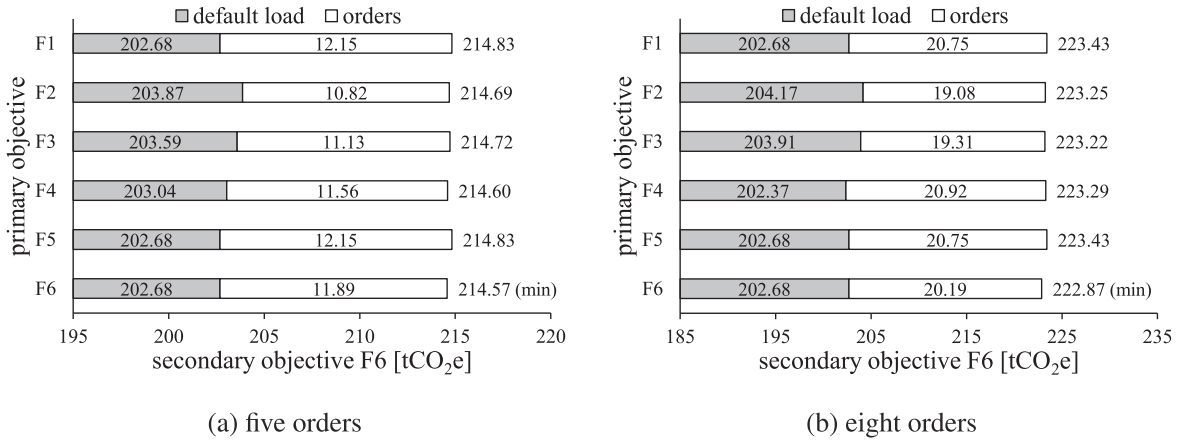



**Fig. 10.** Average results for performance measure "network emissions" (PT-NL).

schemes and it is therefore reasonable to use services with low cost rates under both objectives. In Formula (26a), this can be easily seen as the constant term $\sum_{s\in\mathcal{S}} \underline{c}_s \cdot r_s^{tport}$ is added to performance measure F1. In contrast, objective "order emissions" (F2) minimizes the emissions that are allocated to the orders by either changing the services' allocation scheme or the orders' routing. If the latter is the case, the optimal solutions and network-wide emissions are usually different under objectives F2 and F6. Anyhow, since the total size of the orders is usually small compared to the total size of the services' default load, we only see little differences in this network-wide measure.

$$\min \text{F5} = \text{F1} + \sum_{s\in\mathcal{S}} \underline{c}_s \cdot r_s^{tport} \qquad \text{"network costs"} \tag{26a}$$

$$\min \text{F6} = \text{F2} + \sum_{s\in\mathcal{S}} E_s^{\underline{c}} \qquad \text{"network emissions"} \tag{26b}$$

Fig. 10 shows the network-wide emissions (F6) under primary objectives F1 to F6 for shipping five or eight orders from Portugal to the Netherlands. Thus, the figure shows the minimal network-wide emissions using the optimal value for performance measures F1 to F6. The values are averages of the 25 instances per setting. As mentioned before, the network-wide impact of the chosen objective is limited. For example, the difference between the highest and the lowest value is only 0.26 $tCO_2e$ and 0.56 $tCO_2e$ for the shipping of five and eight orders, respectively. However, the analysis confirms that network-wide emissions are not minimized by minimizing order costs, order emissions, excess emissions, or the number of red orders.

## 5. Conclusion

In this study, we have considered the case of environmentally cautious customers that request to ship orders through an intermodal network in which each order is associated with an emission limit. To reveal if an order is routed such that it stays below its limit, it is necessary to allocate emissions from the used transport services to the orders that use this service. We have considered two emission allocation schemes (egalitarian and payload-based) and let an optimization model select the appropriate scheme for each service. Overall, the problem has been modeled as a **S**ervice **S**election **P**roblem with **T**ransshipment operations, **A**llocation schemes and emission **L**imits in an **E**missions-oriented setting (SSP-TALE). We have solved the problem for cost-, emission- and service-objectives. We have also minimized the number of orders that are routed above their emission limits, which might be a more practical objective for logistics service providers compared to the pure emission-minimizing objective which is often proposed in the vehicle routing literature.

We have tested our approach for a real-world intermodal rail/road network in Europe. In particular, instances of five or eight orders that have to be shipped from Portugal to the Netherlands or from Italy to Denmark have been considered. Experiments have shown the impact of both, the number of orders as well as the country pair, on the results of the considered performance measures. We have shown the interrelations of routing decisions, emission allocation schemes, and emission limits. In particular, solutions where most orders comply with their emission limits are different from solutions where total emissions or total costs are minimized solely. We have also demonstrated that the emissions allocated to an order can increase if a tight emission limit exists for other orders. This is because emission limits impact the selection of allocation schemes and the interdependent routing of orders. Furthermore, our analysis has shown that a payload-based allocation of emissions is often beneficial for orders of small sizes and an egalitarian allocation of emissions is often beneficial for orders of large sizes. Likewise, an egalitarian allocation of emissions is often beneficial if the size of the services' default load is small and a payload-based allocation of emissions is often beneficial if the default load size is large. If both, the orders and the default load, are of medium size, it is often optimal to mix these allocation policies.

Future research may consider the schedules of rail services to analyze the trade-off between transit time and emission limits, and, with this, consider a traditional service-objective. However, powerful heuristics are then required to solve large-scale instances with a large number of orders. In addition, a multi-period planning may yield more possibilities to route orders and, thus, to comply with the orders' emission limits. Another interesting extension could be to explicitly consider customers' willingness to pay for green transportation.

**Acknowledgments**

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## Appendix A. Illustrative example of the SSP-TALE

We present two illustrative examples that demonstrate the interdependencies of emission allocation schemes and emission limits. The same network with the same services is used in both examples. However, the first example considers the case of *one* order with a varied order size and the second example the case of *two* orders with fixed but different order sizes. The considered network consists of only two nodes $a$ and $b$. Each order $o$ is routed from its origin node $a_o = a$ to its destination node $b_o = b$. Thereby, an order $o$ can either use service $s_1$ or service $s_2$ which are two direct, parallel, train services that connect node $a = a_{s_1} = a_{s_2}$ with node $b = b_{s_1} = b_{s_2}$. The distance between the nodes is assumed to be 150 km. Service $s_1$ is a light train and service $s_2$ is a regular train, both with $\underline{m}_{s_1} = \underline{m}_{s_2} = 10$. All other data is used as described in Section 4.1.

We now illustrate with the first example the functioning of the emission allocation schemes and the impact of the order size on the selection of the scheme used per service. For this, we consider the case where a single order $o$ is to be shipped (i.e., $O = \{o\}$) from origin $a_o$ to destination $b_o$ with a load of either $q_o = 2$ TEU ($= 24$ t) or $q_o = 5$ TEU ($= 60$ t). Figs. 11a and 11b report for both order sizes the emissions that are allocated to the default load ($E_s^{\underline{c}}$) and to the considered order $o$ ($E_{os}^{tport}$) using either service $s_1$ or $s_2$ under the egalitarian (EA) or payload-based (PA) allocation scheme. For the order size $q_o = 2$, the lowest emissions allocated to order $o$ are 65 $kgCO_2e$, which results from using service $s_2$ with a payload-based allocation scheme, see Fig. 11a. For the order size $q_o = 5$, using service $s_1$ and an egalitarian allocation effects lowest emissions of 95 $kgCO_2e$, see Fig. 11b. Thus, a payload based allocation is more attractive for small order sizes and an egalitarian allocation of emissions is more attractive for larger order sizes.

For the routing of multiple orders, the amount of emissions allocated to a particular order also depends on the routing of the other orders, which will be shown in the second example. In addition, the second example demonstrates the impact of emission limits on the decisions made as well as such limits' impact on the total network emissions. For this, we route two orders $O = \{o_1, o_2\}$ from node $a$ to node $b$ and, thus, consider the interaction of two jointly routed orders. The quantity of the first order is $q_{o_1} = 2$ TEU and the quantity of the second order is $q_{o_2} = 5$ TEU. Each service $s_1$ or $s_2$ can either be used by $o_1$ or $o_2$ or both, respectively, and emissions on each service can either be allocated with the EA or the PA scheme. Thus, in this example, there exist four routing possibilities and four allocation possibilities. Fig. 12a reports for each such routing-allocation combination the split of total emissions. The numbers in the square bracket atop of the bars serve as index of these combinations. The figure shows that each routing combination leads to different amounts of total greenhouse gases although the differences are small due to the small example size. This amount has its minimum of 2,404 $kgCO_2e$ if both orders use train $s_2$ (solutions [13] to [16]) and its maximum of 2,421 $kgCO_2e$ if both orders use train $s_1$ (solutions [1] to [4]). However, under each routing possibility, the selected emission allocation scheme of the used services decides upon the emissions that are actually allocated to the orders. For example, if both orders use service $s_1$, it is possible to either allocate 93 $kgCO_2e$ to each of the orders (solutions [1] and [2]) or 70 $kgCO_2e$ to order $o_1$ and 175 $kgCO_2e$ to order $o_2$ (solutions [3] and [4]). Fig. 12b shows all possible combinations to distribute emissions among orders $o_1$ and $o_2$. The number(s) in the square brackets refer to the routing-allocation combination from Fig. 12a and the value next to the brackets refers to the total emissions of a solution. Here, it can be seen that the emission minimizing solutions [13] to [16] result from either allocating 128 $kgCO_2e$ to both orders (EA, [13] and [15]) or from allocating 65 $kgCO_2e$ to order $o_1$ and 163 $kgCO_2e$ to order $o_2$ (PA, [14] and [16]).

We now consider the impact of an emission limit $\bar{e}_{o_1} = 80$ for order $o_1$, which is represented by the vertical dotted line in Fig. 12b. If this emission limit is considered as a hard constraint, we are only left with eight routing-allocation solutions, as all other solutions

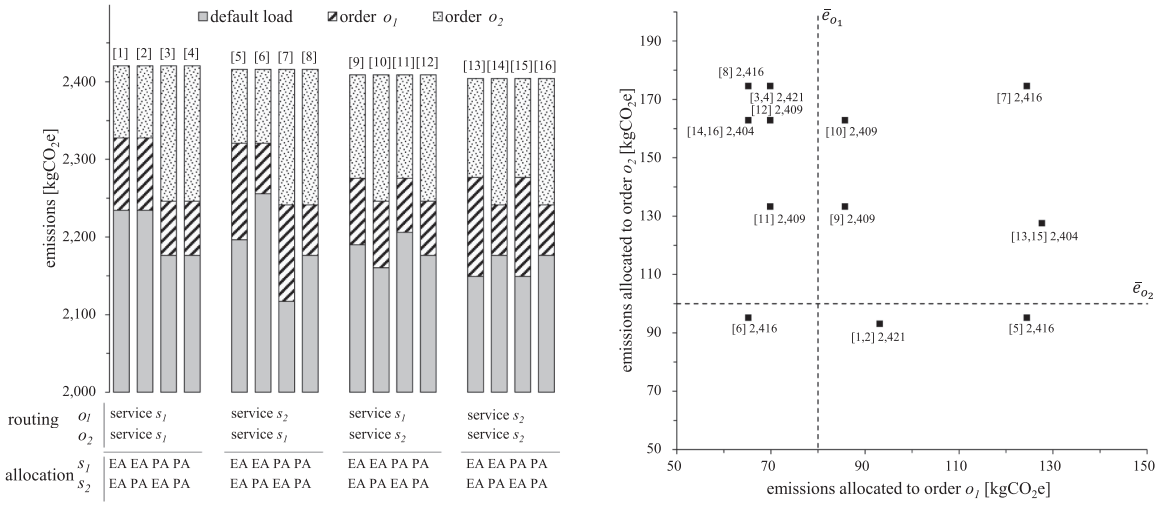
| | allocated emissions [$kgCO_2e$] | | | |
|---|---|---|---|---|
| | egalitarian | | payload-based | |
| service | $E_s^{\underline{c}}$ | $E_{os}^{tport}$ | $E_s^{\underline{c}}$ | $E_{os}^{tport}$ |
| $s_1$ | 857 | 86 | 873 | 70 |
| $s_2$ | 1,244 | 124 | 1,303 | **65** |

(a) $q_o = 2$ TEU

| | allocated emissions [$kgCO_2e$] | | | |
|---|---|---|---|---|
| | egalitarian | | payload-based | |
| service | $E_s^{\underline{c}}$ | $E_{os}^{tport}$ | $E_s^{\underline{c}}$ | $E_{os}^{tport}$ |
| $s_1$ | 952 | **95** | 873 | 175 |
| $s_2$ | 1,333 | 133 | 1,303 | 163 |

(b) $q_o = 5$ TEU

**Fig. 11.** Results for an example with one order of different sizes.




(a) split of total emissions (b) emissions allocated to orders $o_1$ and $o_2$

**Fig. 12.** Solutions for all routing-allocation possibilities in the example with two orders.

would allocate more emissions to order $o_1$ than allowed by the respective limit. Out of the remaining solutions, the lowest emissions allocated to order $o_2$ are 95 $kgCO_2e$, see solution [6]. However, if emission limit $\bar{e}_{o_1}$ is not considered as a hard constraint, solutions [1] and [2] would allocate less than 95 $kgCO_2e$ to order $o_2$. In other words, the emission limit of order $o_1$ restricts attractive routing-allocation solutions that minimize the emissions allocated to order $o_2$. If we additionally consider a hard emission limit of $\bar{e}_{o_2} = 100$ for order $o_2$ (represented by the horizontal dotted line in Fig. 12b), solution [6] remains as the only feasible solution. Here, the allocated emissions comply to both orders' emission limits and the total emissions, consisting of default and order emissions, are 2,416 $kgCO_2e$. At the same time, solution [6] results in 12 $kgCO_2e$ more emissions than solutions [13] to [16] that effect minimum total emissions of 2,404 $kgCO_2e$.

## Appendix B. Implementation of the SSP-TALE

We present in this section the methodology to calculate suitable large numbers (big $M$) and bounds on the emissions allocated to default loads and orders to tighten the SSP-TALE model. For each service $s$, we consider the minimal and maximal emissions than can be allocated to default load $E_s^{\underline{c}}$ and orders $E_{os}^{tport}$ by using either the PA or the EA scheme. Table 5 shows the corresponding values.

If service $s$ uses the PA scheme, the emissions allocated to the default orders are simply calculated by multiplying the services' emission rate $e_s^{tport}$ with default load $\underline{c}_s$ and the emissions allocated to an order $o$ are calculated by multiplying the services' emission rate $e_s^{tport}$ with order load $q_o$. In contrast, if service $s$ uses the EA scheme, the allocated emissions are calculated by considering the number of orders that can use service $s$. Clearly, as the number of orders using a service $s$ depends on the routing in the considered solution, it is not possible to predict the exact emissions that are allocated to each order under the EA scheme. However, it is possible to calculate a lower bound and an upper bound for the emissions per order under the EA scheme. For this, we consider order sizes $q_o$, default load $\underline{c}_s$ and the number of default orders $\underline{m}_s$. Remember that the general procedure of an egalitarian allocation is to divide a service's total emissions by the number of orders that use the service. For example, if only a single order $o$ uses a service $s$ ($x_{os} = 1$ and

**Table 5**
Emissions for the default load and the orders under both allocation schemes.

| | default emissions $E_s^{\underline{c}}$ | order emissions $E_{os}^{tport}$ (if $x_{os} = 1$) |
|---|---|---|
| payload-based allocation (PA) | $e_s^{tport} \cdot \underline{c}_s$ | $e_s^{tport} \cdot q_o$ |
| egalitarian allocation (EA) | | |
| lower bound | $\underline{m}_s \cdot \min\left(\frac{e_s^{tport} \cdot \underline{c}_s}{\underline{m}_s}, \min_{o \in O} LB_{os}\right)$ | $LB_{os}$ as computed by (27) |
| upper bound | $\underline{m}_s \cdot \max\left(\frac{e_s^{tport} \cdot \underline{c}_s}{\underline{m}_s}, \max_{o \in O} UB_{os}\right)$ | $UB_{os}$ as computed by (28) |

$x_{o's} = 0, \forall\, o' \in O | o' \neq o$), the emissions per order are calculated by $\frac{e_s^{tport} \cdot (\underline{c}_s + q_o)}{\underline{m}_s + 1}$. The lower bound $LB_{os}$ and the upper bound $UB_{os}$ for the emissions allocated to order $o$ by using service $s$ under emission allocation scheme EA can be calculated by using Formulas (27) and (28). Here, we consider the $2^{|O|}$ order-combinations of orders $O$, the so-called powerset $P(O)$. For example, the size of the powerset is 32 for the case of 5 orders ($P(O) = \{\emptyset, \{o_1\}, \{o_2\}, \ldots, \{o_1, o_2, o_3, o_4, o_5\}$), where each order $o$ is included in $\frac{|P(O)|}{2} = 16$ sets.

$$LB_{os} = \min_{P' \in P(O) | o \in P'} \left( \frac{e_s^{tport} \cdot \left( \underline{c}_s + \sum_{o' \in P'} q_{o'} \right)}{\underline{m}_s + |P'|} \right), \forall\, o \in O, s \in \mathcal{S} \tag{27}$$

$$UB_{os} = \max_{P' \in P(O) | o \in P'} \left( \frac{e_s^{tport} \cdot \left( \underline{c}_s + \sum_{o' \in P'} q_{o'} \right)}{\underline{m}_s + |P'|} \right), \forall\, o \in O, s \in \mathcal{S} \tag{28}$$

We illustrate the allocation under the EA scheme by considering five orders with order quantities $q_1 = 1$, $q_2 = 3$, $q_3 = 5$, $q_4 = 6$ and $q_5 = 8$ and a service $s$ with emission rate $e_s^{tport} = 1$ $kgCO_2e$, default load $\underline{c}_s = 20$ and $\underline{m}_s = 7$ default orders. Fig. 13a shows the emissions per order for all those sets $P' \in P(O)$ that use service $s$, e.g., $x_{os} = 1\ \forall\, o \in P'$ and $x_{os} = 0\ \forall\, o \in P \backslash P'$. In the figure, the horizontal lines (solid black) are the lower and upper bounds for the emissions per order for using service $s$. If only the most light weight order $o_1$ uses service $s$, the service's total emissions (21 $kgCO_2e$) are equally distributed over $\underline{m}_s + 1 = 8$ orders, resulting in 2.6 $kgCO_2e$ for each of the eight orders (lower bound). In contrast, if orders 3, 4 and 5 use service $s$, the service's total emissions (39 $kgCO_2e$) are equally distributed over $\underline{m}_s + 3 = 10$ orders, resulting in 3.9 $kgCO_2e$ for each of the ten orders (upper bound). Thus, the emissions per order usually increase with the services' load but the extreme values depend on the service's load and number of orders. Fig. 13b shows the emissions per order for all sets $P''$ of powerset $P(O)$ that use service $s$ and include order $o_2$. Here, the bounds (grey dotted lines) that result from sets $P'' \in P(O)$ are more restrictive than the general bounds that result from considering all sets $P' \in P(O)$. In other words, it is reasonable to calculate bounds for each order-service-combination.

Using this information, Formulas (29)–(35) derive the large numbers $M$ for the SSP-TALE model. For $M_o^9$ it is not possible to calculate an appropriate value. However, our experiments indicate that usually all orders can be routed within their emission limit if $p = 1$, e.g., if the emission limit is equivalent to the road-only emissions. We simply use a multiple of the emission limit, although, theoretically, the excess emissions might be higher but we consider such a solution as unacceptable for logistics service providers.

$$M_{os}^1 = M_{os}^2 = \max(e_s^{tport} \cdot q_o, UB_{os}), \forall\, o \in O, s \in \mathcal{S} \tag{29}$$

$$M_s^3 = \max\left( \frac{e_s^{tport} \cdot \underline{c}_s}{\underline{m}_s}, \max_{o \in O} UB_{os} \right), \forall\, s \in \mathcal{S} \tag{30}$$

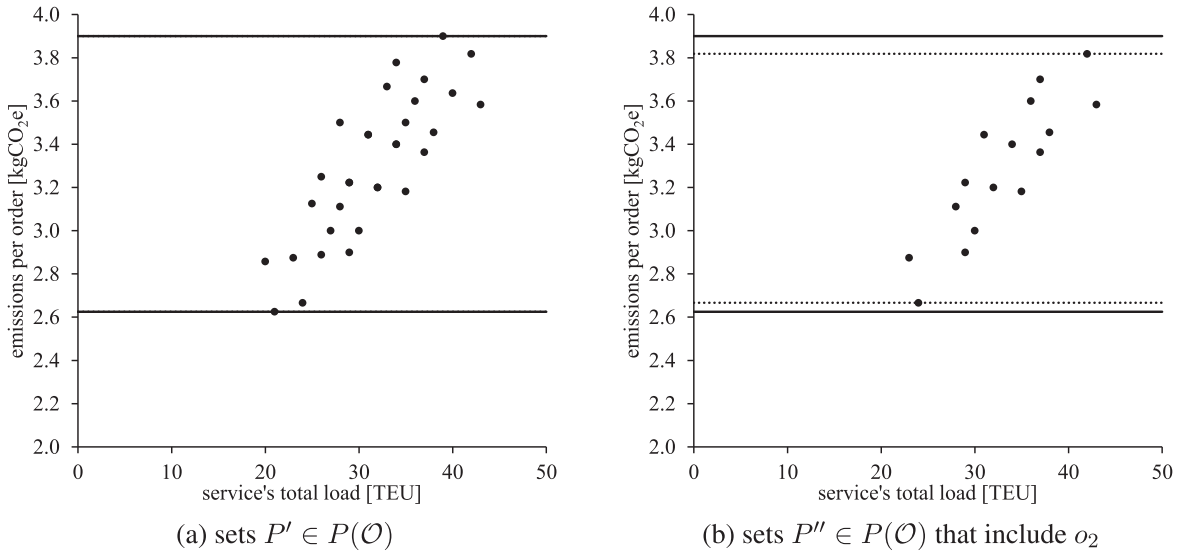



**Fig. 13.** emissions per order from sets of powerset $P(O)$.

$$M_{os}^{4} = e_{s}^{tport} \cdot q_{o}, \forall\, o \in O, s \in \mathcal{S} \tag{31}$$

$$M_{s}^{5} = \underline{m}_{s} \cdot \max_{o \in O} UB_{os}, \forall\, s \in \mathcal{S} \tag{32}$$

$$M_{s}^{6} = e_{s}^{tport} \cdot \underline{c}_{s}, \forall\, s \in \mathcal{S} \tag{33}$$

$$M_{os}^{7} = UB_{os}, \forall\, o \in O, s \in \mathcal{S} \tag{34}$$

$$M_{os}^{8} = e_{s}^{tport} \cdot q_{o}, \forall\, o \in O, s \in \mathcal{S} \tag{35}$$

$$M_{o}^{9} = 100 \cdot \bar{e}_{o}, \forall\, o \in O \tag{36}$$

Finally, Constraints (37) and (38) limit the range of possible values for the emissions allocated to the services' default loads $\underline{c}_s$, Constraints (39) define the maximal emissions for the transshipment of an order $o \in O$ at node $i \in \mathcal{N}$ and Constraints (40) define minimal transport emissions allocated to an order $o \in O$ if service $s \in \mathcal{S}$ is used ($x_{os} = 1$). Overall, Constraints (37)–(40) impose cuts and improve the computational performance significantly, see Section 4.2.

$$E_{s}^{c} \geqslant \underline{m}_{s} \cdot \min\left( \frac{e_{s}^{tport} \cdot \underline{c}_{s}}{\underline{m}_{s}}, \min_{o \in O} LB_{os} \right), \forall\, s \in \mathcal{S} \tag{37}$$

$$E_{s}^{c} \leqslant \underline{m}_{s} \cdot \max\left( \frac{e_{s}^{tport} \cdot \underline{c}_{s}}{\underline{m}_{s}}, \max_{o \in O} UB_{os} \right), \forall\, s \in \mathcal{S} \tag{38}$$

$$E_{oi}^{tship} \leqslant e_{i}^{tship} \cdot q_{o}, \forall\, o \in O, i \in \mathcal{N} \tag{39}$$

$$E_{os}^{tport} \geqslant x_{os} \cdot \min(e_{s}^{tport} \cdot q_{o}, LB_{os}), \forall\, o \in O, s \in \mathcal{S} \tag{40}$$

## Appendix C. Rail network data

The experiments in our study are based on a real-world rail network, see Section 4. This network is described by its nodes, see Tables 6 and 7, and its links, see Table 8. For each rail node, we provide information about the country (ctry), the geographic coordinates (longitude, latitude) and the possibility of rail/road transshipment (tship). In the table, the column "tship" indicates if a

**Table 6**
Rail nodes used in the experiments (1–158).

| id | longitude | latitude | tship | name | id | longitude | latitude | tship | name |
|---|---|---|---|---|---|---|---|---|---|
| Austria (AUT) | | | | | Germany (DEU) | | | | |
| 1 | 14.46404 | 48.221026 | y | Enns | 78 | 10.886843 | 48.36577 | y | Augsburg |
| 2 | 15.474128 | 46.945062 | y | Graz | 79 | 7.040213 | 52.309187 | n | Bad Bentheim |
| 3 | 11.40036 | 47.263281 | n | Innsbruck | 80 | 9.801191 | 50.963577 | y | Bebra |
| 4 | 15.681541 | 47.607642 | y | Mürzzuschlag | 81 | 13.369713 | 52.525051 | y | Berlin |
| 5 | 15.640448 | 46.692045 | y | Spielfeld | 82 | 10.539963 | 52.252608 | y | Braunschweig |
| 6 | 12.165433 | 47.583944 | n | Kufstein | 83 | 8.813076 | 53.082983 | y | Bremen |
| 7 | 13.640441 | 46.534859 | n | Unterhörl | 84 | 8.602726 | 53.534726 | y | Bremerhaven |
| 8 | 16.419189 | 48.169662 | y | Vienna | 85 | 7.013148 | 50.957723 | y | Cologne |
| 9 | 13.821574 | 46.568418 | n | Villach | 86 | 13.708107 | 51.055886 | y | Dresden |
| 10 | 14.061931 | 48.181697 | y | Wels | 87 | 6.77665 | 51.429568 | n | Duisburg |
| | | | | | 88 | 6.836337 | 51.227008 | n | Dusseldorf |
| Belgium (BEL) | | | | | 89 | 9.363728 | 54.814815 | y | Flensburg |
| 11 | 4.427973 | 51.261641 | y | Antwerp | 90 | 14.576137 | 52.321017 | y | Frankfurt (Oder) |
| 12 | 6.017869 | 50.752294 | n | Border BEL/GER | 91 | 8.662662 | 50.10653 | y | Frankfurt am Main |
| 13 | 3.217552 | 51.197724 | n | Bruges | 92 | 12.879519 | 49.332878 | n | Furth im Wald |
| 14 | 4.379132 | 50.878214 | y | Brussels | 93 | 12.318222 | 51.622882 | y | Halle/ Leipzig |
| 15 | 4.44894 | 51.469051 | y | Essen-Grens | 94 | 10.007076 | 53.552812 | y | Hamburg |
| 16 | 3.703306 | 51.037342 | y | Ghent | 95 | 7.809032 | 51.678358 | y | Hamm |
| 17 | 5.61367 | 50.644345 | n | Liège | 96 | 9.741539 | 52.376319 | y | Hanover |
| 18 | 3.839099 | 50.977561 | n | Melle | 97 | 11.923262 | 50.308211 | n | Hof |
| 19 | 5.947037 | 50.722492 | y | Montzen | 98 | 7.664371 | 49.431084 | y | Kaiserslautern |

(*continued on next page*)

**Table 6** (*continued*)

| id | longitude | latitude | tship | name | id | longitude | latitude | tship | name |
|---|---|---|---|---|---|---|---|---|---|
| 20 | 4.862356 | 50.469081 | y | Namur | 99 | 8.402491 | 48.993577 | y | Karlsruhe |
| 21 | 5.714166 | 50.742468 | n | Vise | 100 | 7.806531 | 48.57634 | n | Kehl |
| 22 | 3.183604 | 51.330517 | y | Zeebrugge | 101 | 12.190764 | 47.609938 | n | Kiefersfelden |
| | | | | | 102 | 7.590497 | 50.350254 | n | Koblenz |
| Croatia (HRV) | | | | | 103 | 9.181567 | 48.86196 | y | Kornwestheim |
| 23 | 15.680393 | 45.890551 | y | Dobova | 104 | 8.408974 | 49.535095 | n | Ludwigshafen |
| 24 | 16.815552 | 46.164181 | y | Koprivnica | 105 | 11.627278 | 52.130682 | y | Magdeburg |
| 25 | 14.42843 | 45.330177 | y | Rijeka | 106 | 8.25962 | 50.000914 | n | Mainz |
| 26 | 16.010578 | 45.757174 | y | Zagreb | 107 | 8.442889 | 49.505131 | n | Mannheim |
| | | | | | 108 | 12.087908 | 50.004851 | n | Marktredwitz |
| Czech Republic (CZE) | | | | | 109 | 10.056802 | 53.406594 | n | Maschen |
| 27 | 16.868256 | 48.713414 | n | Breclav | 110 | 11.560487 | 48.140116 | y | Munich |
| 28 | 16.446736 | 49.897183 | y | Ceska Trebova | 111 | 11.082447 | 49.446139 | n | Nuremberg |
| 29 | 14.200856 | 50.773595 | n | Decin | 112 | 8.22318 | 53.143662 | n | Oldenburg |
| 30 | 17.742899 | 49.565266 | n | Hranice | 113 | 13.451574 | 48.563245 | n | Passau |
| 31 | 15.213666 | 50.025209 | n | Kolin | 114 | 12.100189 | 49.012202 | y | Regensburg |
| 32 | 14.492106 | 50.352929 | n | Melnik | 115 | 12.124909 | 54.138555 | y | Rostock |
| 33 | 18.238707 | 49.837242 | y | Ostrava | 116 | 7.003959 | 49.251298 | y | Saarbrücken |
| 34 | 15.757142 | 50.032025 | n | Pardubice | 117 | 14.225089 | 50.859875 | n | Schöna |
| 35 | 13.387889 | 49.743344 | y | Pilsen | 118 | 12.103775 | 49.326648 | n | Schwandorf |
| 36 | 14.501923 | 50.100766 | y | Prague | 119 | 7.562163 | 51.442592 | y | Schwerte |
| 37 | 17.445638 | 49.447083 | y | Prerov | 120 | 8.11518 | 53.518704 | y | Wilhelmshaven |
| | | | | | 121 | 9.936146 | 49.801296 | y | Würzburg |
| Denmark (DNK) | | | | | | | | | |
| 38 | 10.801782 | 55.315005 | y | Nyborg | Hungary (HUN) | | | | |
| 39 | 9.357655 | 54.824635 | n | Padborg | 122 | 19.087919 | 47.468587 | y | Budapest |
| 40 | 9.615722 | 55.545105 | y | Taulov | 123 | 21.252677 | 46.409707 | y | Curtici |
| | | | | | 124 | 16.956545 | 46.249196 | y | Gyékényes |
| France (FRA) | | | | | 125 | 17.095722 | 47.939549 | y | Hegyeshalom |
| 41 | 5.34894 | 45.951993 | n | Amberieu | 126 | 18.115012 | 47.74939 | y | Komarom |
| 42 | 4.838317 | 43.93614 | y | Avignon | 127 | 22.183461 | 48.401387 | y | Záhony |
| 43 | −1.484855 | 43.504819 | n | Bayonne | | | | | |
| 44 | −0.550549 | 44.789377 | y | Bordeaux | Italy (ITA) | | | | |
| 45 | 1.848871 | 50.940848 | y | Calais | 128 | 13.49774 | 43.607495 | y | Ancona |
| 46 | 3.163161 | 42.441462 | n | Cerbere | 129 | 6.710226 | 45.076066 | n | Bardonecchia |
| 47 | 4.843592 | 46.781469 | n | Chalon sur Saone | 130 | 16.870822 | 41.118051 | y | Bari |
| 48 | 5.055213 | 47.322752 | y | Dijon | 131 | 13.656098 | 45.753928 | n | Bivio d'Aurisina |
| 49 | 2.319797 | 51.030104 | y | Dunkirk | 132 | 11.344115 | 44.506286 | y | Bologna |
| 50 | 3.960083 | 49.046564 | n | Epernay | 133 | 13.762419 | 45.706619 | n | Borgo Grotta Gigante |
| 51 | 6.905889 | 49.191647 | n | Forbach | 134 | 11.507424 | 47.005508 | n | Brenner |
| 52 | 1.784988 | 49.285511 | n | Gisors | 135 | 13.343164 | 45.823843 | y | Cervignano A.G. |
| 53 | −1.781844 | 43.35305 | n | Hendaye | 136 | 10.821021 | 45.522415 | y | Domegliara |
| 54 | 4.781583 | 44.799556 | y | La Voulte-sur-Rhöne | 137 | 11.883198 | 44.293985 | y | Faenza |
| 55 | 0.141857 | 49.496632 | y | Le Havre | 138 | 11.251361 | 43.78727 | y | Florence |
| 56 | 3.001735 | 50.634354 | y | Lille | 139 | 8.946657 | 44.406646 | y | Geneo |
| 57 | 4.847635 | 45.727409 | y | Lyon | 140 | 9.813634 | 44.111419 | y | La Spezia |
| 58 | 5.380318 | 43.302687 | y | Marseille | 141 | 10.315923 | 43.564851 | y | Livorno |
| 59 | 6.165476 | 49.096941 | y | Metz (Sablons) | 142 | 9.117494 | 45.511339 | y | Milan |
| 60 | 6.15815 | 49.184697 | n | Metz (Woippy) | 143 | 14.288077 | 40.853414 | y | Napoli |
| 61 | 4.974819 | 43.582786 | n | Miramas | 144 | 8.632783 | 44.630741 | n | Ovada |
| 62 | 6.659711 | 45.193411 | y | Modane | 145 | 11.880506 | 45.418542 | n | Padova |
| 63 | 7.34272 | 47.741867 | y | Mullhouse | 146 | 10.387605 | 43.724167 | y | Pisa |
| 64 | 4.407867 | 43.847584 | n | Nimes | 147 | 11.124314 | 44.981549 | n | Poggio Rusco |
| 65 | 1.904513 | 47.93106 | n | Orleans | 148 | 12.207789 | 44.419118 | y | Ravenna |
| 66 | 2.458354 | 48.896651 | y | Paris (Noisy-le-Sec) | 149 | 12.517239 | 41.892 | y | Roma |
| 67 | 2.637652 | 48.875173 | y | Paris (Vaires-Torcy) | 150 | 15.97007 | 38.488598 | y | Rosarno |
| 68 | 2.877799 | 42.689945 | y | Perpignan | 151 | 17.224245 | 40.483412 | y | Taranto |
| 69 | 0.356004 | 46.625966 | n | Poitiers | 152 | 13.607058 | 46.506519 | y | Tarvisio |
| 70 | 4.884171 | 44.875685 | y | Portes-les-Valence | 153 | 8.862237 | 44.900317 | y | Tortona |
| 71 | 2.460646 | 48.758967 | y | Paris (Valenton) | 154 | 7.679198 | 45.062136 | y | Torino |
| 72 | 1.045719 | 49.444907 | y | Rouen | 155 | 12.258974 | 45.670621 | y | Treviso |
| 73 | 1.539787 | 49.631609 | n | Serqueux | 156 | 13.247456 | 46.092902 | n | Udine |
| 74 | 6.301828 | 45.353571 | n | St Avre | 157 | 12.220498 | 45.484107 | y | Venice |
| 75 | 6.361621 | 45.273882 | n | St Jean de Maurienne | 158 | 10.94991 | 45.427006 | y | Verona |
| 76 | 7.723155 | 48.620542 | y | Strasbourg | | | | | |
| 77 | 0.723724 | 47.385624 | n | Tours | | | | | |

**Table 7**
Rail nodes used in the experiments (159–240).

| id | longitude | latitude | tship | name |
|---|---|---|---|---|
| Luxembourg (LUX) | | | | |
| 159 | 5.824978 | 49.55175 | n | Aubange |
| 160 | 6.107894 | 49.472099 | n | Bettembourg |
| 161 | 6.133952 | 49.590044 | y | Luxembourg |
| Netherlands (NLD) | | | | |
| 162 | 6.655816 | 52.356788 | n | Almelo |
| 163 | 4.899961 | 52.379146 | n | Amsterdam |
| 164 | 4.295976 | 51.495285 | n | Bergen op Zoom |
| 165 | 6.160597 | 52.256971 | n | Deventer |
| 166 | 6.137452 | 51.903058 | y | Emmerich |
| 167 | 3.595915 | 51.444092 | y | Flushing |
| 168 | 6.793398 | 52.262098 | n | Hengelo |
| 169 | 4.459074 | 51.540027 | y | Rosendaal |
| 170 | 4.520059 | 51.89402 | y | Rotterdam |
| 171 | 4.0174 | 51.964998 | y | Rotterdam - Port |
| 172 | 5.109985 | 52.089562 | y | Utrecht |
| Poland (POL) | | | | |
| 173 | 19.004563 | 49.915118 | n | Czechowice-Dziedzice |
| 174 | 22.362249 | 53.824958 | y | Elk |
| 175 | 18.641562 | 54.364692 | y | Gdansk |
| 176 | 18.530088 | 54.521342 | y | Gdynia |
| 177 | 19.017155 | 50.257634 | y | Katowice |
| 178 | 19.429166 | 51.757911 | y | Lodz |
| 179 | 22.389102 | 51.941649 | n | Lukow |
| 180 | 17.92647 | 50.662235 | y | Opele |
| 181 | 16.911403 | 52.401943 | y | Poznan |
| 182 | 15.036678 | 53.331898 | n | Stargard |
| 183 | 14.337132 | 53.897442 | y | Swinoujscie |
| 184 | 14.56616 | 53.412926 | y | Szczecin |
| 185 | 23.600884 | 52.074133 | y | Terespol |
| 186 | 21.051751 | 52.252203 | y | Warsaw |
| 187 | 17.036587 | 51.098655 | y | Wroclaw |
| 188 | 18.971934 | 49.503894 | n | Zwardon |
| Portugal (PRT) | | | | |
| 189 | −7.142046 | 38.895458 | n | Elvas |
| 190 | −8.476715 | 39.461133 | n | Entroncamento |
| 191 | −8.554498 | 38.181846 | n | Grandola |
| 192 | −9.094577 | 38.806741 | y | Lisboa |
| 193 | −8.43132 | 40.33817 | n | Pampilhosa |
| 194 | −8.743508 | 38.634455 | n | Poceirao |
| 195 | −8.684821 | 41.189797 | y | Porto |
| 196 | −8.84305 | 37.939595 | y | Sines |
| 197 | −6.829097 | 40.60615 | n | Vilar Formoso |
| Slovakia (SVK) | | | | |
| 198 | 17.147929 | 48.171346 | y | Bratislava |
| 199 | 22.087799 | 48.419833 | n | Cierna nad Tisou |
| 200 | 18.125549 | 47.768978 | n | Komarno |
| 201 | 18.789116 | 49.479811 | n | Svrcinovec zastávka |
| 202 | 18.809683 | 49.211099 | y | Zilina |
| Slovenia (SVN) | | | | |
| 203 | 13.964778 | 45.681821 | y | Divaca |
| 204 | 14.614992 | 46.059614 | n | Ljubljana |
| 205 | 13.833574 | 45.68758 | n | Orlek |
| 219 | 1.160274 | 41.165484 | n | Constanti |
| 220 | −4.789366 | 37.888424 | y | Cordoba |
| 221 | −6.824784 | 40.59877 | n | Fuentes de Oñoro |
| 222 | −1.165941 | 41.745273 | y | Grisén |
| 223 | −1.788883 | 43.348666 | y | Irun |
| 224 | −0.88809 | 38.792978 | n | La Font de la Figuera |
| 225 | −3.589182 | 38.068974 | n | Lineares Baeza |
| 226 | 0.632775 | 41.620755 | n | Lleida |
| 227 | −3.674798 | 40.386403 | y | Madrid |
| 228 | −4.910277 | 41.316836 | n | Medina del Campo |
| 229 | −6.343821 | 38.92173 | n | Mérida |
| 230 | −4.63277 | 37.580747 | n | Montilla |
| 231 | −1.027458 | 38.007272 | n | Murcia |
| 232 | −1.661343 | 42.824984 | y | Pamplona |
| 233 | 3.157742 | 42.424373 | y | Portbou |
| 234 | −5.974816 | 37.391586 | y | Seville |
| 235 | 1.253371 | 41.111766 | y | Tarragona |
| 236 | −0.370571 | 39.441337 | y | Valencia |
| 237 | −4.726878 | 41.642059 | n | Valladolid |
| 238 | −0.891337 | 41.643813 | n | Zaragoza |
| Switzerland (CHE) | | | | |
| 239 | 7.596036 | 47.544822 | y | Basel |
| 240 | 9.033657 | 45.832351 | y | Chiasso |

(*continued on next page*)

**Table 7** (*continued*)

| id | longitude | latitude | tship | name | id | longitude | latitude | tship | name |
|---|---|---|---|---|---|---|---|---|---|
| 206 | 15.662149 | 46.395269 | n | Pragersko | | | | | |
| 207 | 14.338531 | 45.934199 | y | Vrhnika | | | | | |
| 208 | 15.170675 | 46.085288 | n | Zidani Most | | | | | |
| Spain (ESP) | | | | | | | | | |
| 209 | −5.449048 | 36.126428 | y | Algeciras | | | | | |
| 210 | −2.455037 | 36.835094 | y | Almeria | | | | | |
| 211 | −4.683008 | 40.6571 | n | Avila | | | | | |
| 212 | −6.981698 | 38.890678 | n | Badajoz | | | | | |
| 213 | 2.192621 | 41.385915 | y | Barcelona | | | | | |
| 214 | −3.106599 | 43.363535 | y | Bilbao | | | | | |
| 215 | −4.696158 | 37.043883 | n | Bobadilla | | | | | |
| 216 | −3.666404 | 42.371886 | n | Burgos | | | | | |
| 217 | −0.974878 | 37.605154 | y | Cartagena | | | | | |
| 218 | −3.913527 | 38.985061 | n | Ciudad Real | | | | | |

**Table 8**
Rail links used in the experiments.

| id [–] | from [–] | to [–] | distance [km] | id [–] | from [–] | to [–] | distance [km] | id [–] | from [–] | to [–] | distance [km] | id [–] | from [–] | to [–] | distance [km] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 113 | 10 | 82.7 | 72 | 48 | 59 | 255.3 | 143 | 72 | 73 | 67.5 | 214 | 93 | 81 | 134.3 |
| 2 | 10 | 1 | 38.5 | 73 | 59 | 160 | 48.1 | 144 | 73 | 52 | 50.6 | 215 | 97 | 108 | 42.1 |
| 3 | 1 | 8 | 183.7 | 74 | 160 | 161 | 15.4 | 145 | 52 | 67 | 88.3 | 216 | 118 | 108 | 96.0 |
| 4 | 8 | 4 | 118.0 | 75 | 161 | 159 | 24.3 | 146 | 66 | 67 | 14.3 | 217 | 114 | 118 | 42.5 |
| 5 | 4 | 2 | 109.5 | 76 | 45 | 56 | 105.3 | 147 | 67 | 50 | 294.4 | 218 | 114 | 113 | 118.9 |
| 6 | 2 | 5 | 33.8 | 77 | 49 | 56 | 83.2 | 148 | 50 | 59 | 210.6 | 219 | 134 | 3 | 36.6 |
| 7 | 2 | 9 | 221.2 | 78 | 56 | 66 | 231.7 | 149 | 60 | 59 | 12.9 | 220 | 101 | 6 | 3.7 |
| 8 | 9 | 7 | 15.5 | 79 | 56 | 14 | 124.3 | 150 | 59 | 51 | 70.0 | 221 | 3 | 6 | 72.5 |
| 9 | 8 | 27 | 80.2 | 80 | 16 | 13 | 38.4 | 151 | 59 | 76 | 149.9 | 222 | 101 | 110 | 89.4 |
| 10 | 8 | 198 | 66.2 | 81 | 13 | 22 | 15.3 | 152 | 51 | 116 | 11.5 | 223 | 86 | 117 | 52.9 |
| 11 | 198 | 202 | 219.7 | 82 | 16 | 11 | 71.3 | 153 | 116 | 98 | 60.6 | 224 | 93 | 86 | 146.5 |
| 12 | 202 | 199 | 331.7 | 83 | 16 | 18 | 11.6 | 154 | 98 | 104 | 77.3 | 225 | 81 | 86 | 176.8 |
| 13 | 198 | 125 | 46.2 | 84 | 18 | 14 | 48.1 | 155 | 98 | 107 | 76.0 | 226 | 81 | 90 | 94.8 |
| 14 | 125 | 200 | 93.7 | 85 | 11 | 14 | 47.5 | 156 | 46 | 233 | 2.0 | 227 | 118 | 92 | 73.8 |
| 15 | 126 | 200 | 7.7 | 86 | 20 | 14 | 60.3 | 157 | 233 | 213 | 167.2 | 228 | 76 | 100 | 12.8 |
| 16 | 126 | 122 | 94.1 | 87 | 17 | 14 | 104.0 | 158 | 213 | 235 | 92.4 | 229 | 100 | 239 | 144.6 |
| 17 | 127 | 122 | 330.2 | 88 | 11 | 15 | 23.7 | 159 | 235 | 238 | 262.9 | 230 | 117 | 29 | 11.5 |
| 18 | 123 | 122 | 222.3 | 89 | 17 | 21 | 16.8 | 160 | 235 | 226 | 103.9 | 231 | 29 | 32 | 85.0 |
| 19 | 124 | 122 | 251.5 | 90 | 11 | 21 | 128.2 | 161 | 226 | 238 | 138.1 | 232 | 92 | 35 | 74.5 |
| 20 | 124 | 26 | 103.6 | 91 | 21 | 19 | 17.9 | 162 | 235 | 236 | 257.3 | 233 | 36 | 31 | 56.5 |
| 21 | 26 | 25 | 230.4 | 92 | 19 | 12 | 7.6 | 163 | 238 | 222 | 43.1 | 234 | 31 | 34 | 43.2 |
| 22 | 26 | 23 | 44.4 | 93 | 20 | 159 | 168.7 | 164 | 236 | 224 | 99.1 | 235 | 34 | 28 | 59.7 |
| 23 | 23 | 208 | 51.2 | 94 | 15 | 169 | 8.2 | 165 | 224 | 231 | 144.8 | 236 | 28 | 37 | 108.9 |
| 24 | 208 | 204 | 56.8 | 95 | 169 | 164 | 13.4 | 166 | 231 | 217 | 54.5 | 237 | 27 | 37 | 105.5 |
| 25 | 204 | 207 | 35.9 | 96 | 164 | 167 | 61.8 | 167 | 215 | 234 | 152.3 | 238 | 37 | 30 | 28.4 |
| 26 | 207 | 203 | 75.9 | 97 | 169 | 170 | 52.0 | 168 | 234 | 220 | 126.8 | 239 | 30 | 33 | 53.8 |
| 27 | 208 | 206 | 73.5 | 98 | 170 | 171 | 47.8 | 169 | 220 | 218 | 172.8 | 240 | 202 | 201 | 39.7 |
| 28 | 206 | 5 | 35.3 | 99 | 170 | 172 | 61.2 | 170 | 218 | 227 | 172.6 | 241 | 201 | 33 | 74.6 |
| 29 | 203 | 205 | 12.9 | 100 | 170 | 163 | 75.1 | 171 | 79 | 96 | 209.2 | 242 | 201 | 188 | 25.3 |
| 30 | 205 | 133 | 7.4 | 101 | 172 | 163 | 39.5 | 172 | 96 | 109 | 159.3 | 243 | 188 | 173 | 69.5 |
| 31 | 133 | 131 | 11.7 | 102 | 172 | 165 | 79.2 | 173 | 109 | 94 | 19.9 | 244 | 90 | 181 | 173.1 |
| 32 | 131 | 135 | 29.0 | 103 | 165 | 162 | 38.6 | 174 | 96 | 80 | 180.6 | 245 | 181 | 182 | 172.0 |

(*continued on next page*)

**Table 8** (*continued*)

| id [–] | from [–] | to [–] | distance [km] | id [–] | from [–] | to [–] | distance [km] | id [–] | from [–] | to [–] | distance [km] | id [–] | from [–] | to [–] | distance [km] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 33 | 7 | 152 | 6.4 | 104 | 162 | 168 | 14.5 | 175 | 80 | 121 | 152.6 | 246 | 182 | 184 | 36.4 |
| 34 | 152 | 156 | 82.1 | 105 | 168 | 79 | 18.2 | 176 | 80 | 91 | 158.7 | 247 | 184 | 183 | 105.6 |
| 35 | 156 | 135 | 40.4 | 106 | 172 | 166 | 75.7 | 177 | 96 | 83 | 122.2 | 248 | 181 | 186 | 305.4 |
| 36 | 135 | 157 | 100.8 | 107 | 196 | 191 | 77.4 | 178 | 83 | 84 | 62.6 | 249 | 181 | 178 | 256.6 |
| 37 | 155 | 157 | 22.3 | 108 | 191 | 194 | 57.1 | 179 | 83 | 112 | 44.2 | 250 | 178 | 186 | 140.2 |
| 38 | 155 | 156 | 108.1 | 109 | 194 | 192 | 66.7 | 180 | 112 | 120 | 52.5 | 251 | 181 | 187 | 184.2 |
| 39 | 157 | 145 | 28.0 | 110 | 194 | 190 | 139.5 | 181 | 83 | 94 | 116.7 | 252 | 187 | 180 | 81.6 |
| 40 | 145 | 158 | 85.3 | 111 | 192 | 190 | 95.9 | 182 | 94 | 89 | 182.9 | 253 | 180 | 177 | 97.5 |
| 41 | 158 | 147 | 53.1 | 112 | 190 | 189 | 158.9 | 183 | 94 | 81 | 286.2 | 254 | 177 | 186 | 301.4 |
| 42 | 147 | 132 | 59.5 | 113 | 189 | 212 | 16.0 | 184 | 81 | 115 | 227.5 | 255 | 186 | 179 | 115.9 |
| 43 | 158 | 136 | 17.7 | 114 | 190 | 193 | 125.3 | 185 | 81 | 105 | 145.6 | 256 | 186 | 175 | 331.3 |
| 44 | 136 | 134 | 215.7 | 115 | 193 | 195 | 125.8 | 186 | 105 | 82 | 82.8 | 257 | 175 | 176 | 19.9 |
| 45 | 158 | 142 | 152.2 | 116 | 193 | 197 | 201.6 | 187 | 82 | 96 | 66.6 | 258 | 173 | 177 | 43.3 |
| 46 | 142 | 139 | 157.1 | 117 | 197 | 221 | 1.2 | 188 | 166 | 87 | 80.1 | 259 | 178 | 179 | 229.8 |
| 47 | 142 | 240 | 55.8 | 118 | 212 | 229 | 59.1 | 189 | 87 | 88 | 40.3 | 260 | 179 | 185 | 92.2 |
| 48 | 144 | 153 | 54.3 | 119 | 229 | 227 | 405.9 | 190 | 88 | 85 | 34.7 | 261 | 186 | 174 | 274.7 |
| 49 | 144 | 139 | 45.6 | 120 | 209 | 215 | 180.1 | 191 | 85 | 119 | 81.3 | 262 | 89 | 39 | 2.7 |
| 50 | 154 | 129 | 88.7 | 121 | 215 | 230 | 74.4 | 192 | 119 | 95 | 34.2 | 263 | 40 | 39 | 101.3 |
| 51 | 129 | 62 | 18.8 | 122 | 230 | 220 | 48.8 | 193 | 95 | 96 | 176.1 | 264 | 40 | 38 | 90.5 |
| 52 | 132 | 137 | 48.9 | 123 | 221 | 228 | 199.9 | 194 | 12 | 85 | 83.2 | 265 | 24 | 124 | 14.6 |
| 53 | 137 | 148 | 33.4 | 124 | 228 | 211 | 87.8 | 195 | 85 | 102 | 92.8 | 266 | 41 | 62 | 184.1 |
| 54 | 137 | 128 | 154.8 | 125 | 211 | 227 | 132.0 | 196 | 102 | 106 | 100.0 | 267 | 41 | 57 | 52.6 |
| 55 | 132 | 138 | 92.2 | 126 | 228 | 237 | 46.2 | 197 | 106 | 91 | 37.3 | 268 | 54 | 68 | 351.7 |
| 56 | 138 | 146 | 79.9 | 127 | 227 | 237 | 189.4 | 198 | 91 | 104 | 100.7 | 269 | 42 | 61 | 68.1 |
| 57 | 146 | 141 | 19.3 | 128 | 237 | 216 | 124.8 | 199 | 91 | 121 | 135.4 | 270 | 64 | 68 | 216.7 |
| 58 | 146 | 140 | 72.3 | 129 | 216 | 223 | 264.1 | 200 | 121 | 111 | 126.9 | 271 | 70 | 68 | 357.6 |
| 59 | 140 | 139 | 84.0 | 130 | 232 | 223 | 155.6 | 201 | 104 | 99 | 73.3 | 272 | 68 | 233 | 45.7 |
| 60 | 138 | 149 | 260.3 | 131 | 222 | 232 | 157.2 | 202 | 99 | 103 | 80.3 | 273 | 74 | 62 | 38.0 |
| 61 | 149 | 143 | 236.0 | 132 | 227 | 222 | 285.6 | 203 | 99 | 100 | 74.5 | 274 | 75 | 62 | 27.2 |
| 62 | 143 | 130 | 292.1 | 133 | 214 | 223 | 303.2 | 204 | 76 | 63 | 111.8 | 275 | 153 | 154 | 111.8 |
| 63 | 130 | 151 | 104.5 | 134 | 223 | 53 | 0.9 | 205 | 48 | 63 | 204.2 | 276 | 219 | 213 | 118.9 |
| 64 | 143 | 150 | 406.6 | 135 | 53 | 43 | 36.8 | 206 | 63 | 239 | 34.5 | 277 | 225 | 210 | 240.6 |
| 65 | 239 | 240 | 311.0 | 136 | 43 | 44 | 198.2 | 207 | 103 | 78 | 183.7 | 278 | 225 | 220 | 127.0 |
| 66 | 62 | 57 | 235.2 | 137 | 44 | 69 | 257.2 | 208 | 78 | 110 | 61.9 | 279 | 107 | 91 | 90.9 |
| 67 | 57 | 42 | 232.8 | 138 | 69 | 77 | 95.9 | 209 | 78 | 111 | 138.6 | 280 | 68 | 64 | 216.7 |
| 68 | 61 | 58 | 54.8 | 139 | 77 | 65 | 112.6 | 210 | 110 | 114 | 138.0 | 281 | 64 | 42 | 45.2 |
| 69 | 68 | 46 | 43.6 | 140 | 65 | 71 | 110.5 | 211 | 111 | 114 | 100.4 | 282 | 107 | 104 | 13.5 |
| 70 | 57 | 47 | 128.3 | 141 | 71 | 67 | 24.7 | 212 | 93 | 97 | 200.7 | 283 | 142 | 136 | 168.7 |
| 71 | 47 | 48 | 74.0 | 142 | 55 | 72 | 113.1 | 213 | 93 | 105 | 86.2 | 284 | 24 | 26 | 88.9 |

node qualifies for rail/road transshipment (“y”), or not (“n”). Nodes that do not qualify for transshipment are artificial rail nodes and used for the routing of trains through the network. For example, such nodes are used to calculate the correct distance (in km) between the actual stops of a train. Each rail link connects two rail nodes as shown in Table 8. The distance for each link is calculated by raildar.fr. We refer to Heinold and Meisel (2018, 2019) for further information on this.

In this rail network, we model light and regular trains as services, see Table 9. Here, “node sequence” lists all nodes that are passed by the train while traveling from the service’s origin to the service’s destination. The train stops for rail/road transshipment at all nodes that are classified as transshipment nodes, e.g., “tship” = “y”. These nodes are listed in column “stop sequence”.

**Table 9**
Train services used in the experiments.

| id | type | node sequence | stop sequence | id | type | node sequence | stop sequence |
|---|---|---|---|---|---|---|---|
| 1 | light train | 195 193 190 192 | 195 192 | 54 | light train | 146 141 | 146 141 |
| 2 | light train | 192 190 193 195 | 192 195 | 55 | light train | 128 137 132 | 128 137 132 |
| 3 | light train | 192 194 191 196 | 192 196 | 56 | light train | 132 137 128 | 132 137 128 |
| 4 | light train | 196 191 194 192 | 196 192 | 57 | light train | 148 137 | 148 137 |
| 5 | light train | 196 191 194 190 189 212 229 227 | 196 227 | 58 | light train | 137 148 | 137 148 |
| 6 | light train | 227 229 212 189 190 194 191 196 | 227 196 | 59 | light train | 48 63 239 100 99 107 91 | 48 63 239 99 91 |
| 7 | light train | 195 193 197 221 228 211 227 | 195 227 | 60 | light train | 91 107 99 100 239 63 48 | 91 99 239 63 48 |
| 8 | light train | 227 211 228 221 197 193 195 | 227 195 | 61 | light train | 99 103 78 110 | 99 103 78 110 |
| 9 | light train | 209 215 230 220 218 227 | 209 220 227 | 62 | light train | 110 78 103 99 | 110 78 103 99 |
| 10 | light train | 227 218 220 230 215 209 | 227 220 209 | 63 | light train | 120 112 83 94 | 120 83 94 |
| 11 | light train | 234 220 225 210 | 234 220 210 | 64 | light train | 94 83 112 120 | 94 83 120 |
| 12 | light train | 210 225 220 234 | 210 220 234 | 65 | light train | 38 40 39 89 94 | 38 40 89 94 |
| 13 | light train | 217 231 224 236 235 213 | 217 236 235 213 | 66 | light train | 94 89 39 40 38 | 94 89 40 38 |
| 14 | light train | 213 235 236 224 231 217 | 213 235 236 217 | 67 | light train | 115 81 | 115 81 |
| 15 | light train | 214 223 | 214 223 | 68 | light train | 81 115 | 81 115 |
| 16 | light train | 223 214 | 223 214 | 69 | light train | 183 184 182 181 | 183 184 181 |
| 17 | light train | 235 219 226 238 222 232 223 | 235 222 232 223 | 70 | light train | 181 182 184 183 | 181 184 183 |
| 18 | light train | 223 232 222 238 226 219 235 | 223 232 222 235 | 71 | light train | 181 187 180 177 | 181 187 180 177 |
| 19 | light train | 66 67 71 | 66 67 71 | 72 | light train | 177 180 187 181 | 177 180 187 181 |
| 20 | light train | 71 67 66 | 71 67 66 | 73 | light train | 185 179 186 178 181 | 185 186 178 181 |
| 21 | light train | 58 61 42 | 58 42 | 74 | light train | 181 178 186 179 185 | 181 178 186 185 |
| 22 | light train | 42 61 58 | 42 58 | 75 | light train | 174 186 | 174 186 |
| 23 | light train | 68 54 | 68 54 | 76 | light train | 186 174 | 186 174 |
| 24 | light train | 54 68 | 54 68 | 77 | light train | 35 92 118 114 | 35 114 |
| 25 | light train | 68 70 | 68 70 | 78 | light train | 114 118 92 35 | 114 35 |
| 26 | light train | 70 68 | 70 68 | 79 | light train | 114 113 10 1 8 198 125 200 126 122 123 | 114 10 1 8 198 125 126 122 123 |
| 27 | light train | 55 72 73 52 66 | 55 72 66 | 80 | light train | 123 122 126 200 125 198 8 1 10 113 114 | 123 122 126 125 198 8 1 10 114 |
| 28 | light train | 66 52 73 72 55 | 66 72 55 | 81 | light train | 127 122 124 24 26 25 | 127 122 124 24 26 25 |
| 29 | light train | 67 50 59 | 67 59 | 82 | light train | 25 26 24 124 122 127 | 25 26 24 124 122 127 |
| 30 | light train | 59 50 67 | 59 67 | 83 | light train | 26 23 208 206 5 2 | 26 23 5 2 |
| 31 | light train | 45 56 | 45 56 | 84 | light train | 2 5 206 208 23 26 | 2 5 23 26 |
| 32 | light train | 56 45 | 56 45 | 85 | light train | 26 23 208 204 207 203 205 133 131 135 157 145 158 | 26 23 207 203 135 157 158 |
| 33 | light train | 49 56 | 49 56 | 86 | light train | 158 145 157 135 131 133 205 203 207 204 208 23 26 | 158 157 135 203 207 23 26 |
| 34 | light train | 56 49 | 56 49 | 87 | light train | 91 121 | 91 121 |
| 35 | light train | 22 13 16 18 14 17 21 19 12 85 | 22 16 14 19 85 | 88 | light train | 121 91 | 121 91 |
| 36 | light train | 85 12 19 21 17 14 18 16 13 22 | 85 19 14 16 22 | 89 | regular train | 227 222 238 235 213 233 46 68 64 42 70 57 47 48 59 | 227 222 235 213 233 68 42 70 57 48 59 |
| 37 | light train | 167 164 169 | 167 169 | 90 | regular train | 59 48 47 57 70 42 64 68 46 233 213 235 238 222 227 | 59 48 57 70 42 68 233 213 235 222 227 |
| 38 | light train | 169 164 167 | 169 167 | 91 | regular train | 227 237 216 223 53 43 44 69 77 65 71 | 227 223 44 71 |
| 39 | light train | 171 170 163 172 166 87 88 85 102 106 91 | 171 170 172 166 85 91 | 92 | regular train | 71 65 77 69 44 43 53 223 216 237 227 | 71 44 223 227 |
| 40 | light train | 91 106 102 85 88 87 166 172 163 170 171 | 91 85 166 172 170 171 | 93 | regular train | 66 56 14 11 15 169 170 172 165 162 168 79 96 109 94 | 66 56 14 11 15 169 170 172 96 94 |
| 41 | light train | 59 60 160 161 159 20 14 | 59 161 20 14 | 94 | regular train | 94 109 96 79 168 162 165 172 170 169 15 11 14 56 66 | 94 96 172 170 169 15 11 14 56 66 |
| 42 | light train | 14 20 159 161 160 60 59 | 14 20 161 59 | 95 | regular train | 150 143 149 138 132 147 158 136 134 3 6 101 110 | 150 143 149 138 132 158 136 110 |
| 43 | light train | 59 51 116 98 104 91 | 59 116 98 91 | 96 | regular train | 110 101 6 3 134 136 158 147 132 138 149 143 150 | 110 136 158 132 138 149 143 150 |
| 44 | light train | 91 104 98 116 51 59 | 91 98 116 59 | 97 | regular train | 110 78 111 121 80 96 109 94 | 110 78 121 80 96 94 |
| 45 | light train | 85 119 95 96 | 85 119 95 96 | 98 | regular train | 94 109 96 80 121 111 78 110 | 94 96 80 121 78 110 |

(*continued on next page*)

**Table 9** (*continued*)

| id | type | node sequence | stop sequence | id | type | node sequence | stop sequence |
|---|---|---|---|---|---|---|---|
| 46 | light train | 96 95 119 85 | 96 95 119 85 | 99 | regular train | 84 83 96 82 105 93 86 117 29 32 36 31 34 28 37 30 33 201 202 199 127 | 84 83 96 82 105 93 86 36 28 37 33 202 127 |
| 47 | light train | 143 130 151 | 143 130 151 | 100 | regular train | 127 199 202 201 33 30 37 28 34 31 36 32 29 117 86 93 105 82 96 83 84 | 127 202 33 37 28 36 86 93 105 82 96 83 84 |
| 48 | light train | 151 130 143 | 151 130 143 | 101 | regular train | 94 81 90 181 186 | 94 81 90 181 186 |
| 49 | light train | 57 41 74 75 62 129 154 153 144 139 140 146 138 | 57 62 154 153 139 140 146 138 | 102 | regular train | 186 181 90 81 94 | 186 181 90 81 94 |
| 50 | light train | 138 146 140 139 144 153 154 129 62 75 74 41 57 | 138 146 140 139 153 154 62 57 | 103 | regular train | 176 175 186 177 173 188 201 202 201 33 30 37 27 8 4 2 9 7 152 156 155 157 145 158 | 176 175 186 177 202 33 37 8 4 2 152 155 157 158 |
| 51 | light train | 59 76 63 239 240 142 136 | 59 76 63 239 240 142 136 | 104 | regular train | 158 145 157 155 156 152 7 9 2 4 8 27 37 30 33 201 202 201 188 173 177 186 175 176 | 158 157 155 152 2 4 8 37 33 202 177 186 175 176 |
| 52 | light train | 136 142 240 239 63 76 59 | 136 142 240 239 63 76 59 | 105 | regular train | 110 114 118 108 97 93 81 | 110 114 93 81 |
| 53 | light train | 141 146 | 141 146 | 106 | regular train | 81 93 97 108 118 114 110 | 81 93 114 110 |

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